

Multimodal models

CS 5624: Natural Language Processing

Fall 2026

<https://tuvllms.github.io/nlp-fall-2026/>

Tu Vu

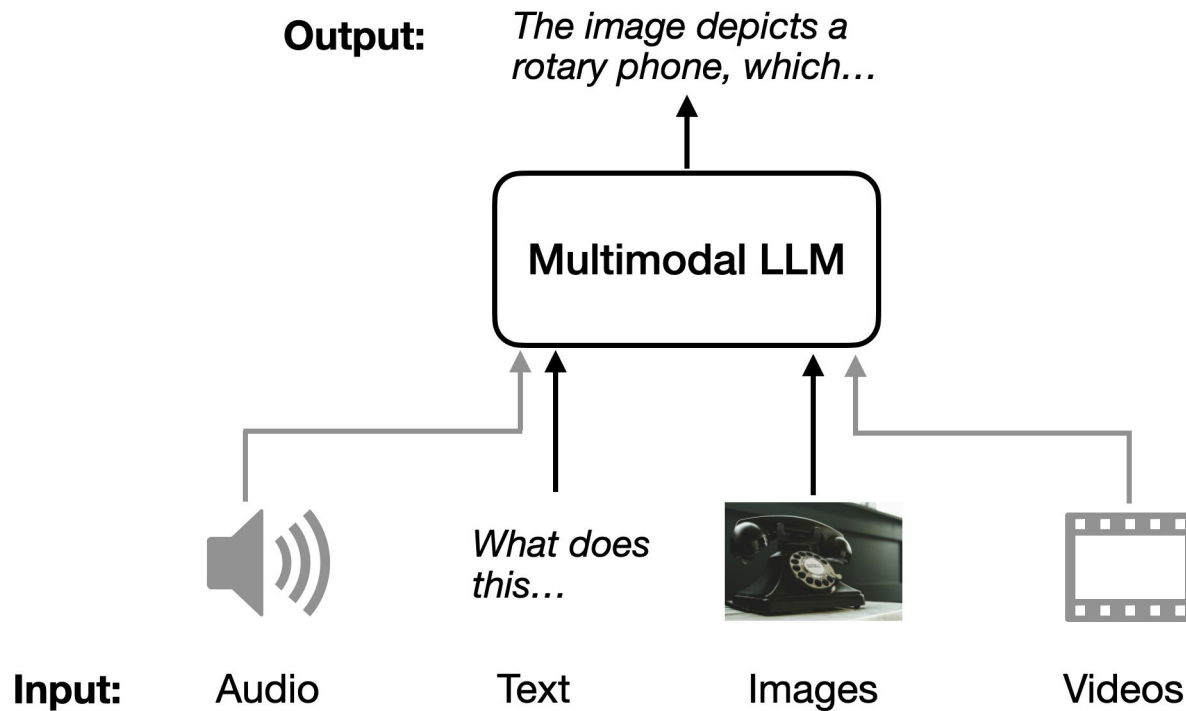


Logistics

- Final projects
 - Project proposal
 - Template & samples available soon
- Quiz1 due 9/28 (next Monday)
- HW1 due 10/5

AI news

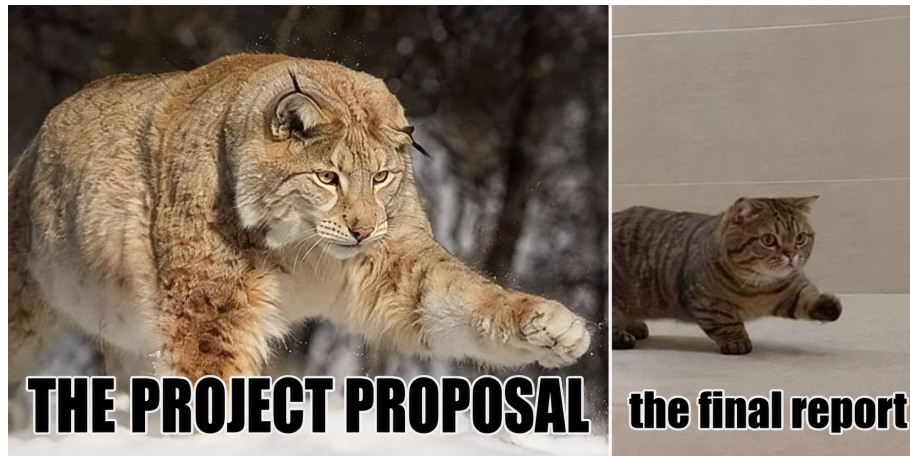
A multimodal model can process and generate information across multiple modalities



Use cases of multimodal models

- Text-to-Image
- Image Edit
- Text-to-Video
- Image-to-Video
- Video Edit
- ...

Use cases of multimodal models



Explain the meme

The joke is about the huge gap between **how ambitious a project sounds at the beginning** and **what actually gets accomplished by the deadline**.

- **"THE PROJECT PROPOSAL"** is represented by a lynx: huge, powerful, dramatic, and charging through the snow. The proposal promises an impressive project with ambitious goals, sophisticated methods, and substantial results.
- **"the final report"** is represented by a small house cat awkwardly trotting along. By the end, the actual work is much smaller and less impressive than what the proposal envisioned.

The typography reinforces the joke too: **THE PROJECT PROPOSAL** is enormous and all caps, whereas **the final report** is comparatively small.

So the basic progression is:

"Here is the groundbreaking project we're going to do." → months pass → "Well, here's what we managed to do."

Use cases of multimodal models (cont'd)

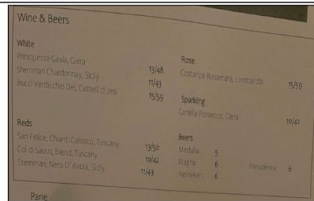
(a) User:  { "smartphone": 1, "teddy bear": 1 }  { "cat": 3 }  { "book": 3, "vase": 1, "glass": 1 } 	MM1-30B (Ours): { "dog": 2, "frisbee": 1 }
(b) User:  Red circle: "no parking anytime"  Red circle: "Raffaello"  Red circle: "Rue Saint-Paul"  Red circle:	MM1-30B (Ours): "Hyde Park"
(c) User:  furniture: bed frame, weight: 50 and 150 pounds (23 to 68 kg)  furniture: sofa, weight: 100 to 200 pounds (45 to 91 kg)  furniture: stove, weight: 150 to 300 pounds (68 to 136 kg)  furniture:	MM1-30B (Ours): refrigerator, weight: 200 to 300 pounds (91 to 136 kg)
(d) User:   total: $1 + 3 = 4$   total: $6 + 4 = 10$   total: $4 + 1 = 5$	MM1-30B (Ours): total: $4 + 1 = 5$

Fig. 1: MM1 can perform in-context predictions thanks to its large-scale multimodal pre-training. This allows MM1 to (a) count objects and follow custom formatting, (b) refer to parts of the images and perform OCR, (c) demonstrate common-sense and word knowledge about everyday objects, and (d) perform basic math functions. Images are from the COCO 2014 validation set [72].

Use cases of multimodal models (cont'd)

User:

Photo: <img1> Menu: <img2> How much should I pay for all the beer on the table according to the price on the menu?



MM1-30B-Chat (Ours): 12

Emu-Chat-37B: According to the menu, the price for all the beer on the table is **15.99**.

LLaVA-NeXT-34B: The image shows a menu with beer prices, but it's not entirely clear which specific beers are on the table. However, I can provide you with an estimate based on the visible prices. From the menu, it appears that the prices for beer are as follows: Magna: 5, Medalla: 5, Heineken: 6 Presidente: 6

Assuming that the beers on the table are Magna, Medalla, Heineken, and Presidente, and there are two of each, you would pay:

2 x Magna: $2 \times 5 = 10$, 2 x Medalla: $2 \times 5 = 10$, 2 x Heineken: $2 \times 6 = 12$, 2 x Presidente: $2 \times 6 = 12$ Total: $10 + 10 + 12 + 12 = 44$

Please note that this is an estimate based on the visible prices and the assumption that there are two of each type of beer. The actual cost may vary if the beers on the table are not the ones listed on the menu or if there are different quantities.

User: Explain why.

MM1-30B-Chat (Ours): There are 2 beers on the table, and each beer costs 6 according to the price on the menu. So, $2 \times 6 = 12$.

Emu-Chat-37B: The price for all the beer on the table is 15.99 because each beer has a price of 10.99, and there are a total of five beers on the table. When you multiply the number of beers (5) by the price per beer (10.99), you get a total cost of 54.95 (rounded up to 15.99).

LLaVA-NeXT-34B: The image shows a menu with beer prices, but it's not entirely clear which specific beers are on the table. However, I can provide you with an estimate based on the visible prices. From the menu, it appears that the prices for beer are as follows: (...)

Fig. 2: MM1 can follow instructions and reason across images. Example and images from VILA [71]; VILA answers correctly when prompted with chain-of-thought.

[MM1: Methods, Analysis & Insights from Multimodal LLM Pre-training](#)

Models

- GPT-Image
- MAI-Image
- Grok-Image
- Reve
- Muse-image
- Gemini-Image (Nano Banana)
- Seedream
- Qwen-image
- ...
- Gemini-Omni
- Flux
- Grok-Imagine-Video
- Dreamina-seedance
- ...

<https://arena.ai/leaderboard/>
<https://artificialanalysis.ai/>

The resolution curse

384x384

What type of shop is in the image?

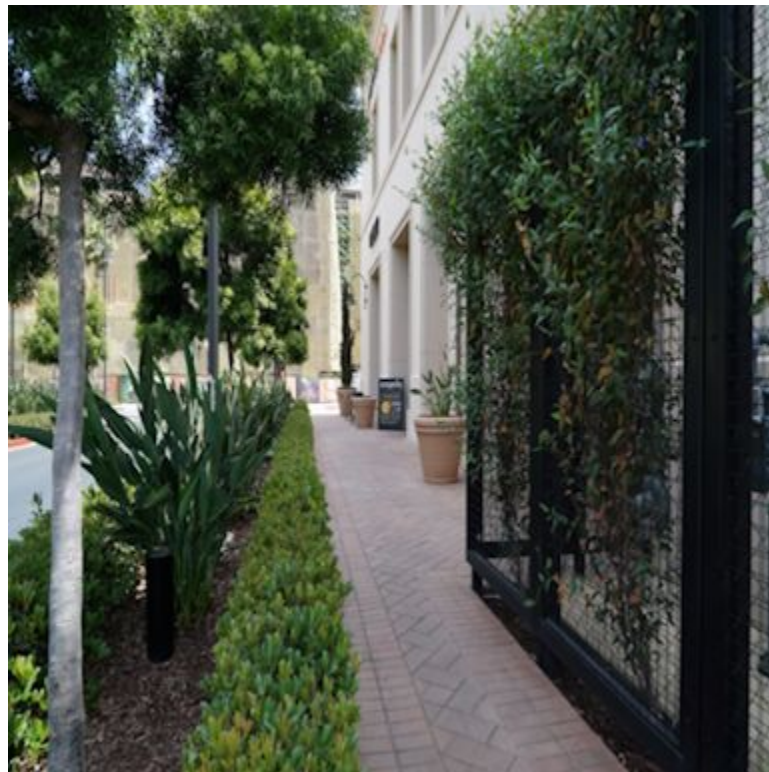


<https://huggingface.co/blog/visheratin/vlm-resolution-curse>

The resolution curse

384x384

What type of shop is in the image?



<https://huggingface.co/blog/visheratin/vlm-resolution-curse>

Grayscale images



0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

Color images



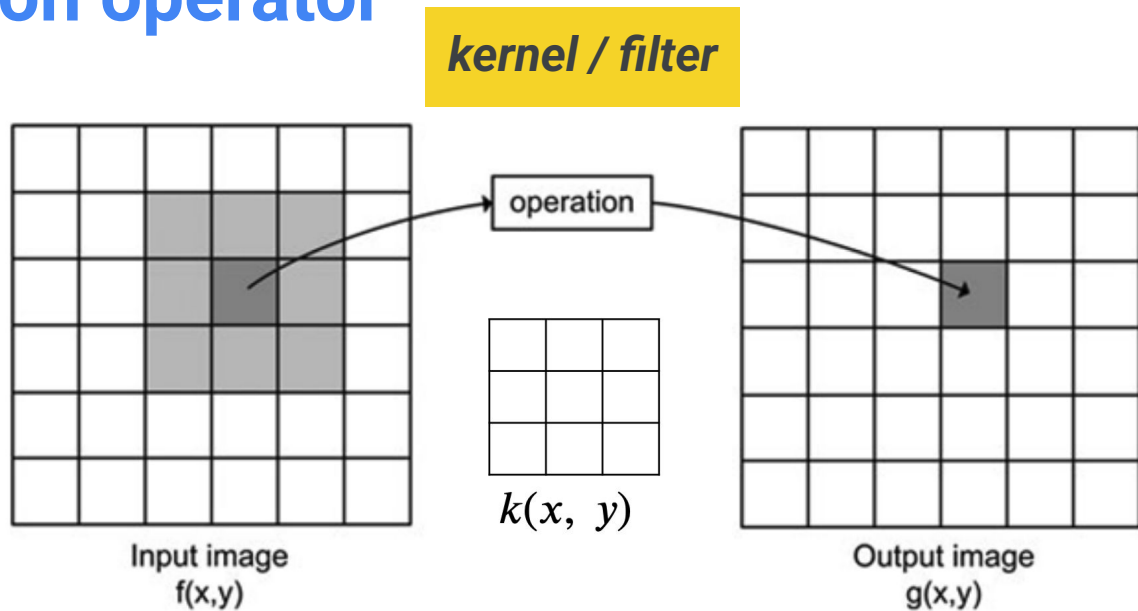
0	3	2	5	1	7	6	0	8
0	3	2	5	1	7	6	0	8
0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

channel x height x width

Channels are usually RGB: Red, Green, and Blue

Convolutional neural network (CNN)

Convolution operator



$$g(x, y) = \sum_v \sum_u k(u, v) f(x - u, y - v)$$

Input image

4	5	7	6	6
3	2	8	0	7
6	7	7	1	5
3	0	1	1	1
4	3	2	1	7

*

Weights

*

0	0	0
1	0	1
0	0	0



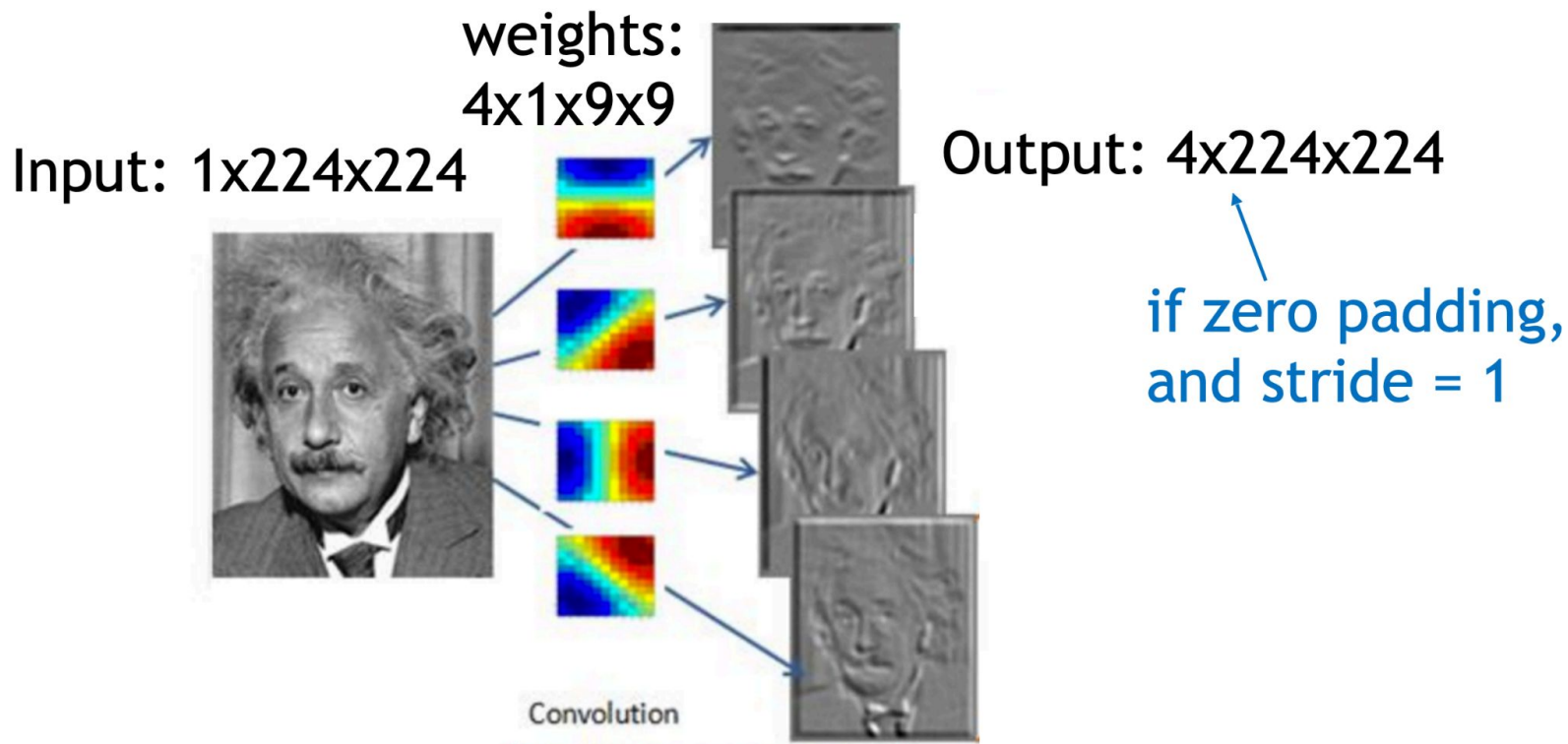
Output image

	11	2	15	
	13	8	12	
	?			

Demo

- <https://setosa.io/ev/image-kernels/>

Convolutional layer (with 4 filters)



ImageNet Classification with Deep Convolutional Neural Networks

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Ilya Sutskever

University of Toronto

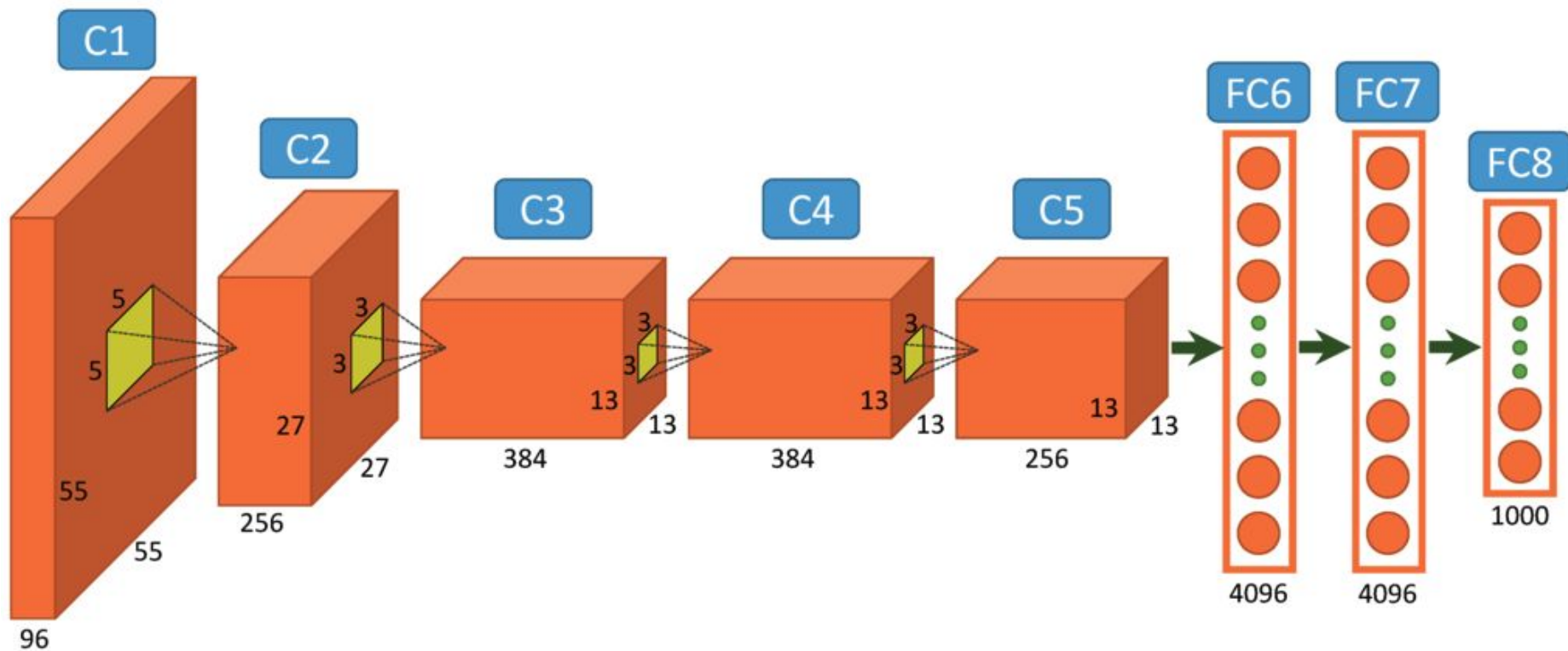
ilya@cs.utoronto.ca

Geoffrey E. Hinton

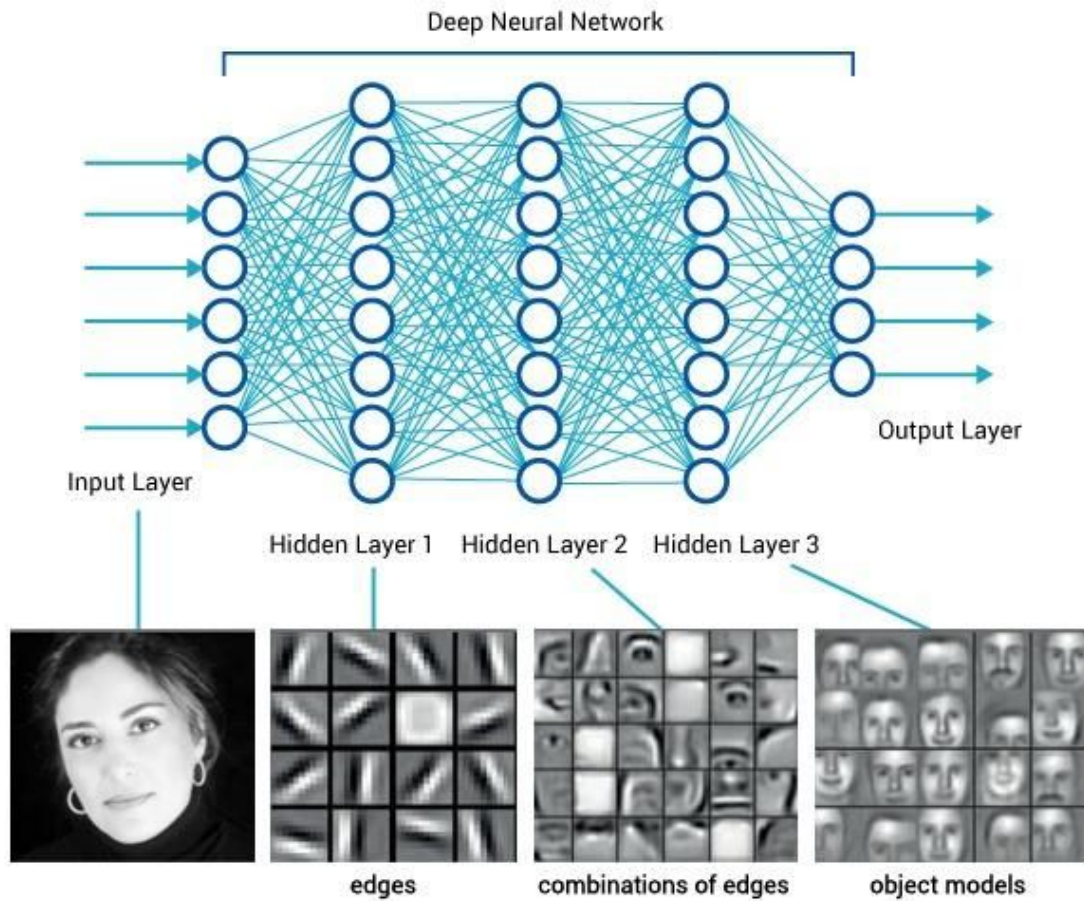
University of Toronto

hinton@cs.utoronto.ca

AlexNet



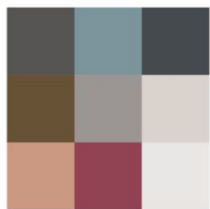
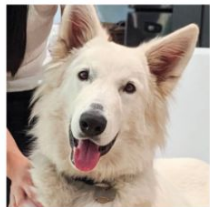
AlexNet (cont'd)



Vision Transformers

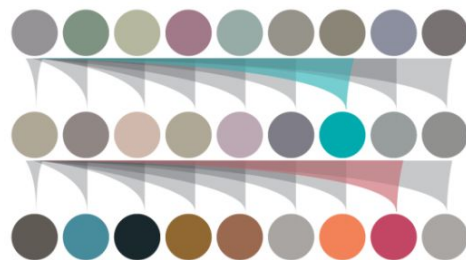
Next-pixel prediction?

I



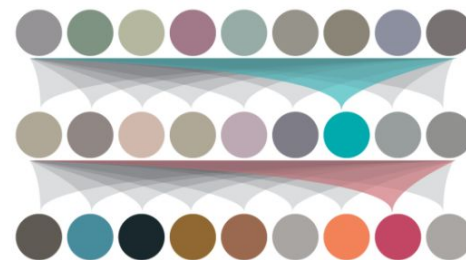
2

(a) Autoregressive



Target

(b) BERT



Target

AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

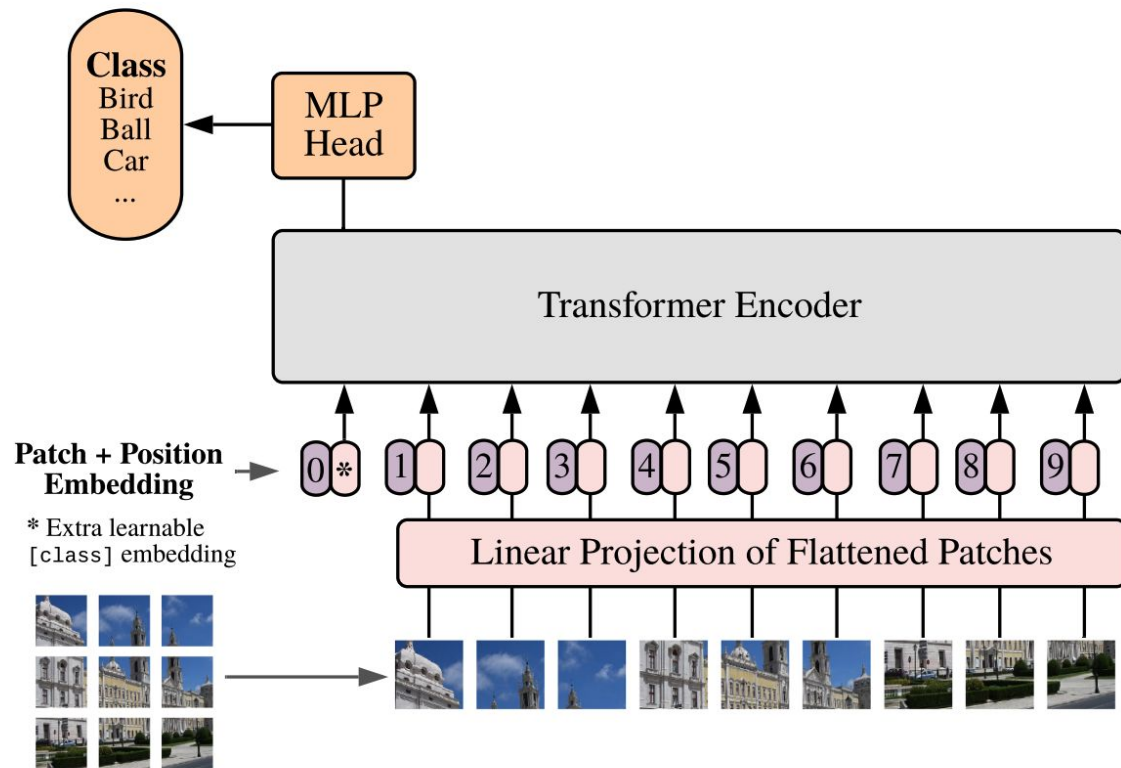
**Alexey Dosovitskiy^{*,†}, Lucas Beyer^{*}, Alexander Kolesnikov^{*}, Dirk Weissenborn^{*},
Xiaohua Zhai^{*}, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer,
Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby^{*,†}**

^{*}equal technical contribution, [†]equal advising

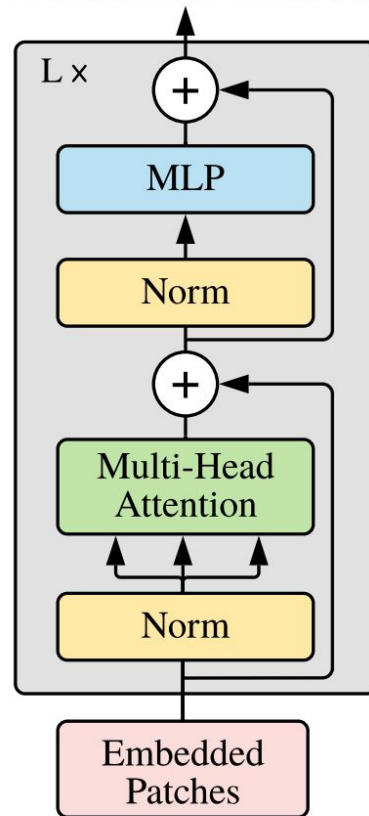
Google Research, Brain Team

{adosovitskiy, neilhoulby}@google.com

Vision Transformer (ViT)



Transformer Encoder



BERT recap

softmax

linear



Image created by
Gemini

[CLS]



Multi-head Self-attention
(unmasked)



[CLS]

the

movie

was

good

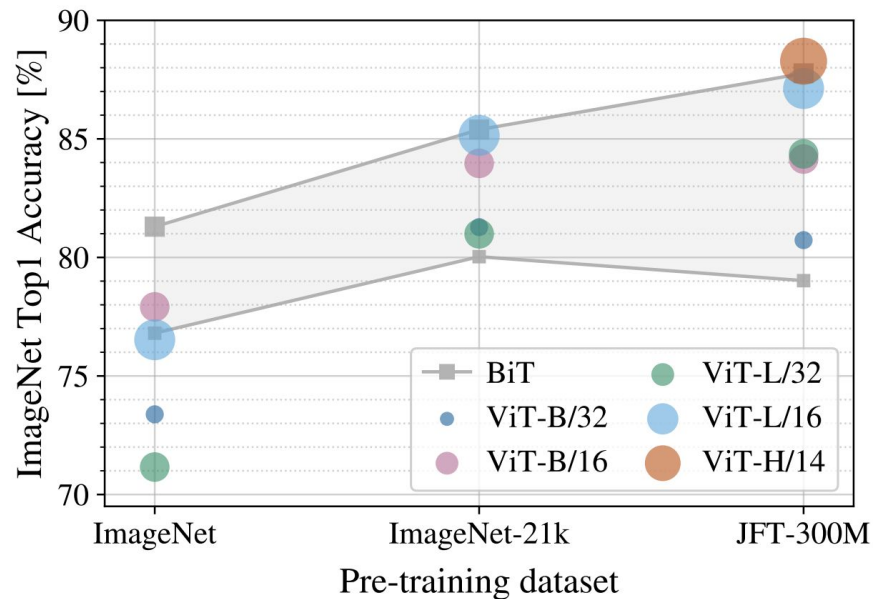
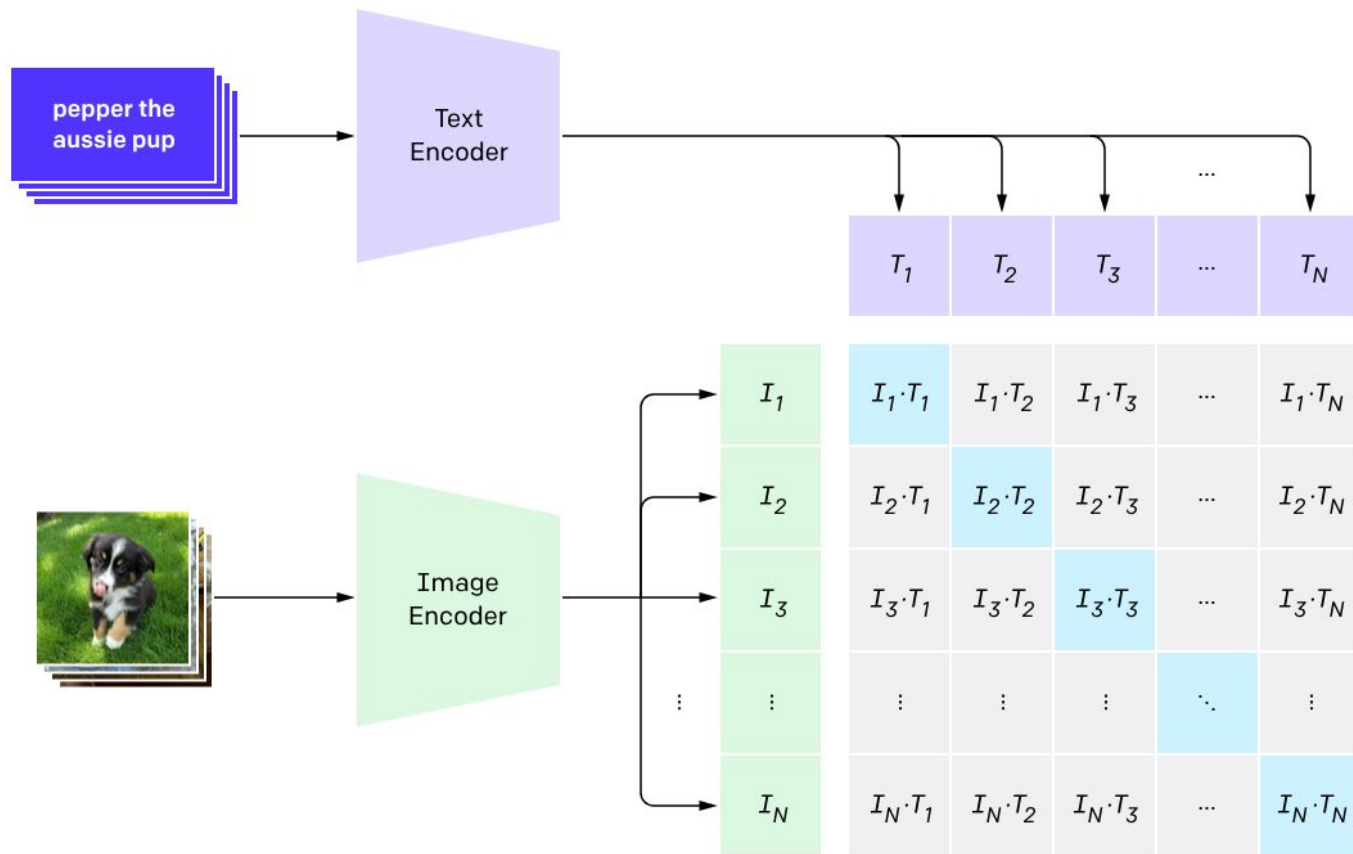


Figure 3: Transfer to ImageNet. While large ViT models perform worse than BiT ResNets (shaded area) when pre-trained on small datasets, they shine when pre-trained on larger datasets. Similarly, larger ViT variants overtake smaller ones as the dataset grows.

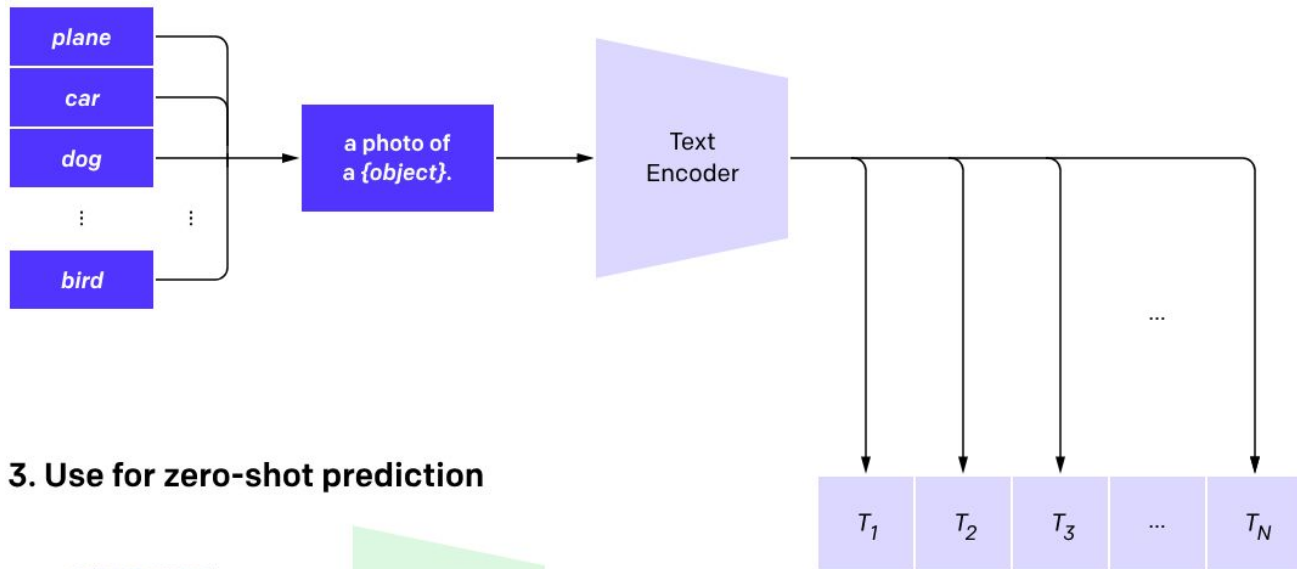
OpenAI's CLIP

1. Contrastive pre-training

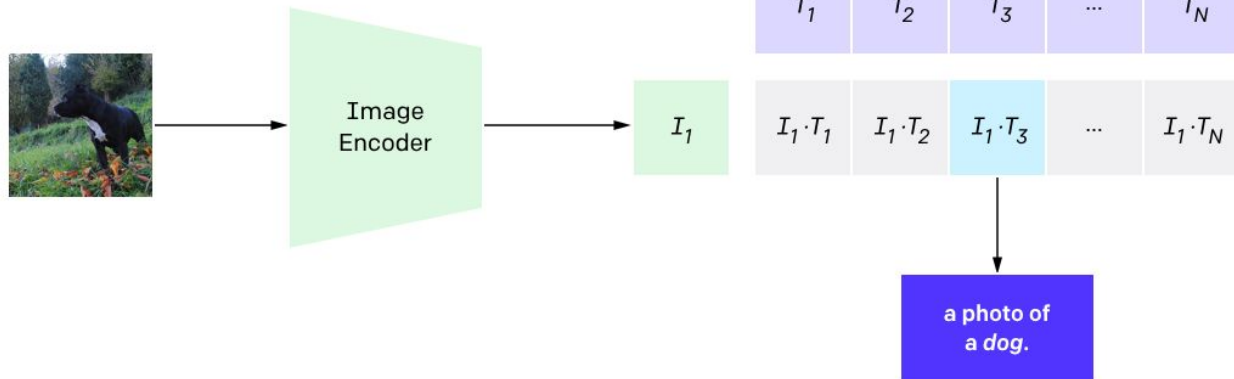


OpenAI's CLIP (cont'd)

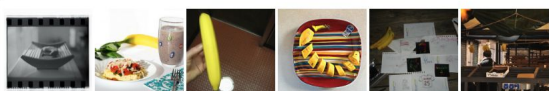
2. Create dataset classifier from label text



3. Use for zero-shot prediction



OpenAI's CLIP (cont'd)

Dataset	ImageNet	
	ResNet101	CLIP ViT-L
	76.2%	76.2%
ImageNet		
	64.3%	70.1%
ImageNet V2		
	37.7%	88.9%
ImageNet Rendition		
	32.6%	72.3%
ObjectNet		
	25.2%	60.2%
ImageNet Sketch		
	2.7%	77.1%
ImageNet Adversarial		

<https://openai.com/index/clip/>

Contrastive Language Models



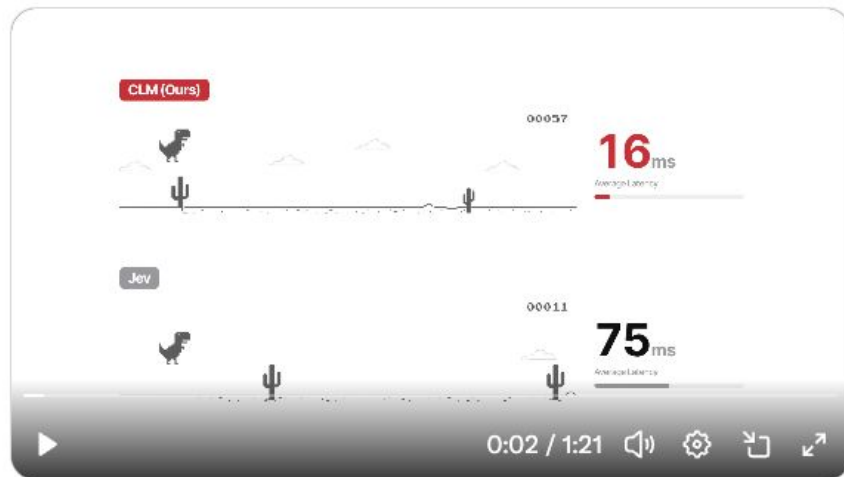
Jacky Kwok @jackyk02 · 17h



Introducing Contrastive Language Model (CLM): an ultra-fast System One Model trained with a contrastive learning objective that connects states and actions.

CLM-8B is pre-trained on internet-scale data and delivers up to 9× faster inference than Jev ⚡ while achieving comparable

[Show more](#)



139

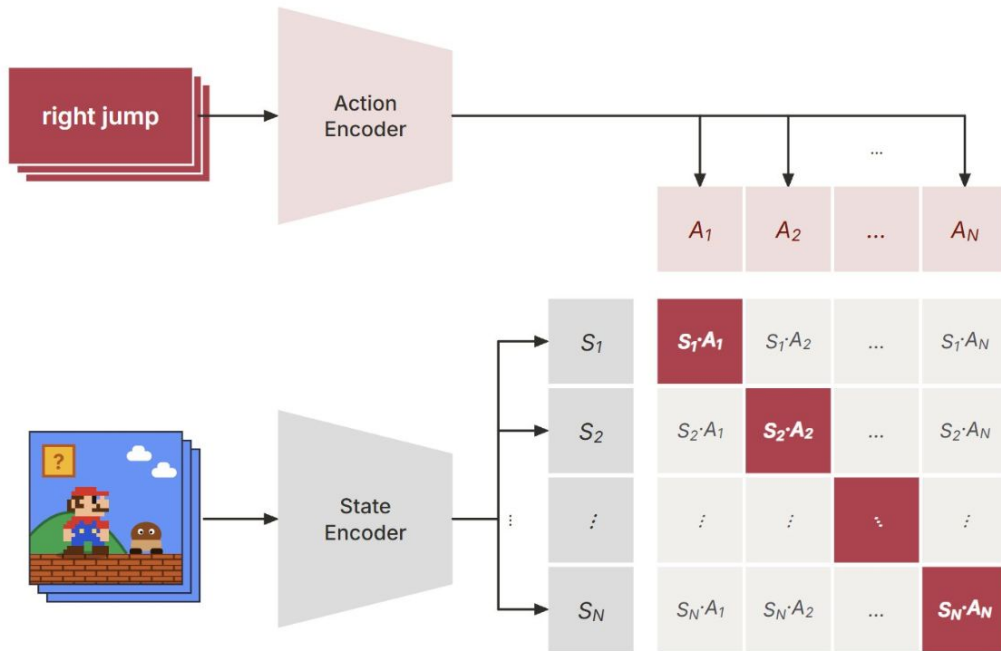
540

3.9K

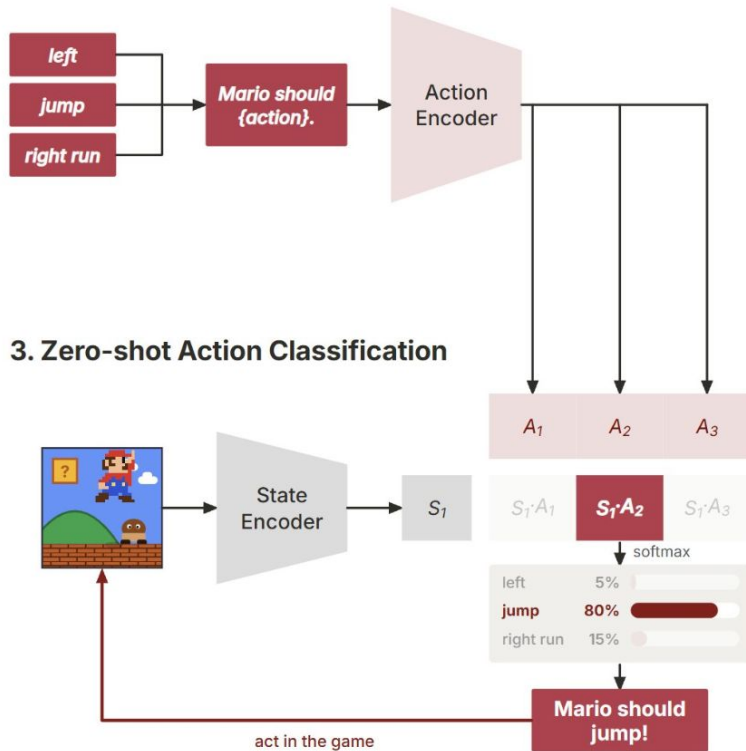
561K



1. Contrastive State-Action Pre-training



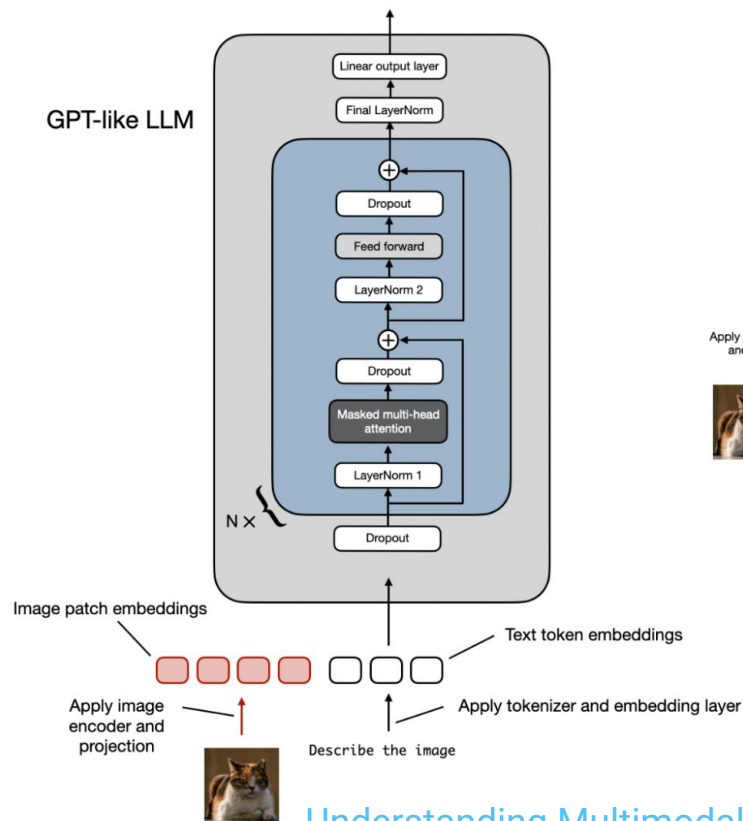
2. Create Action Candidates



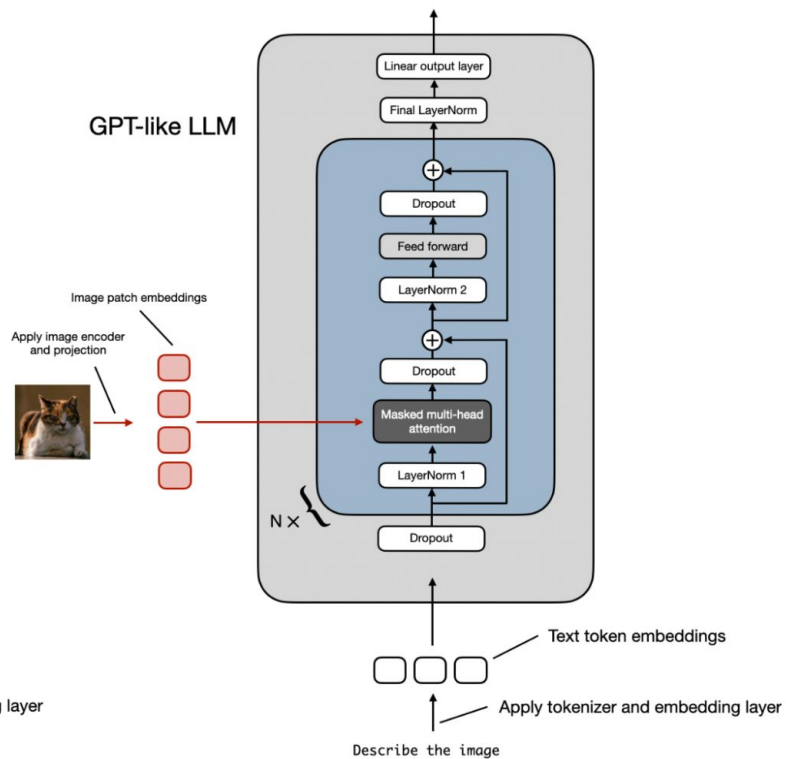
3. Zero-shot Action Classification

Common approaches to building multimodal models

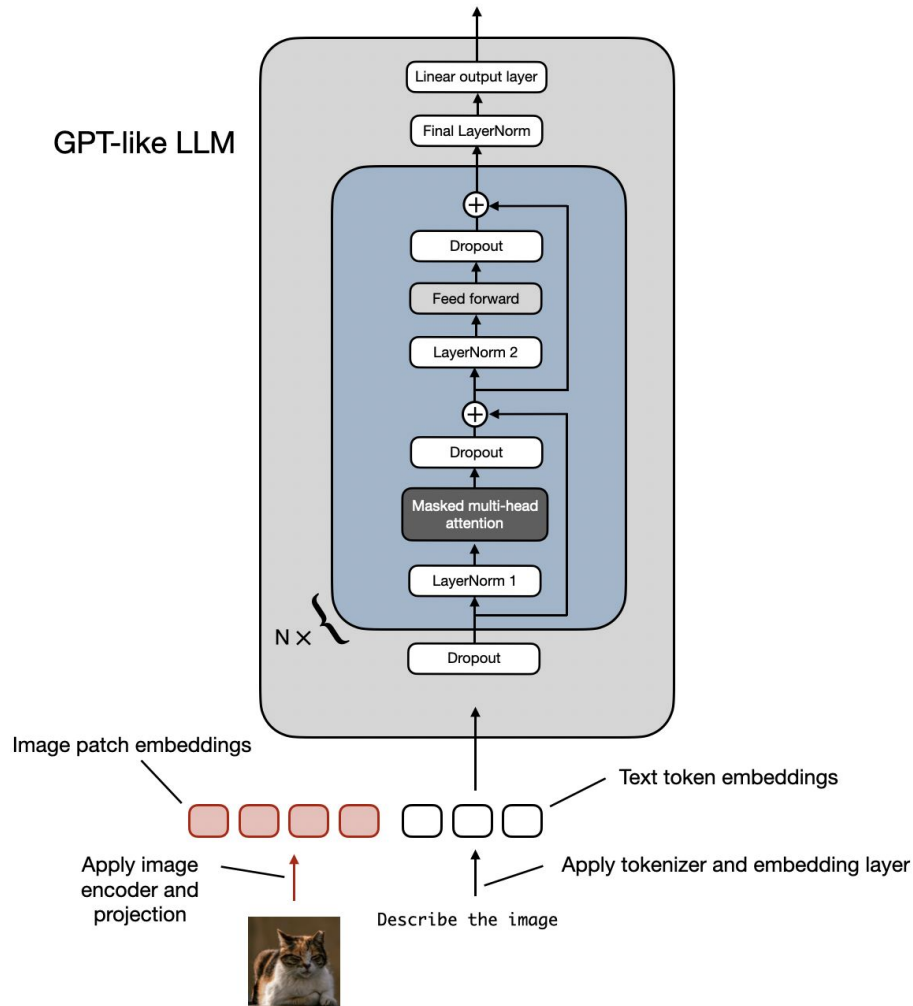
Method A: Unified Embedding Decoder Architecture



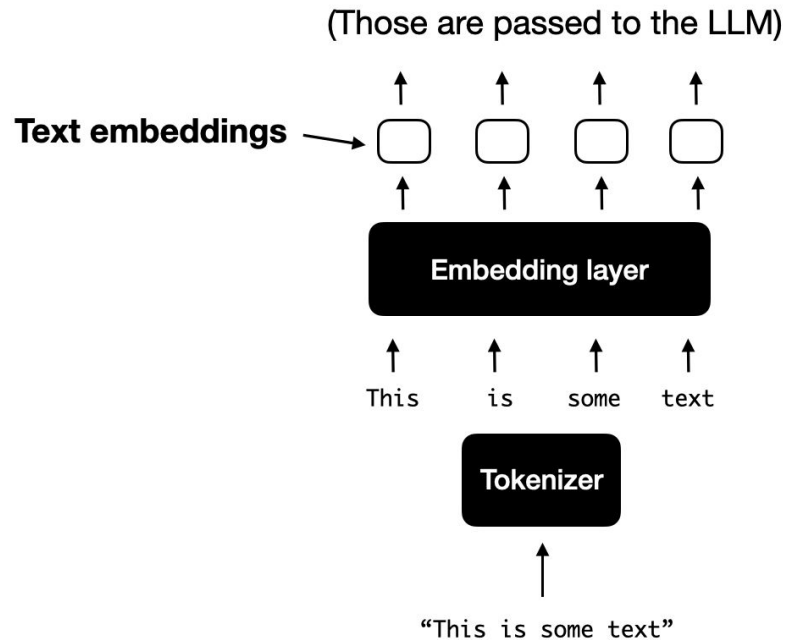
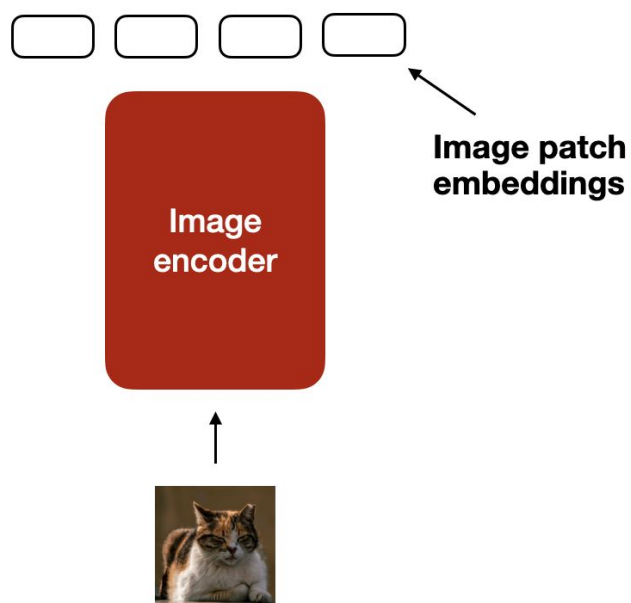
Method B: Cross-Modality Attention Architecture



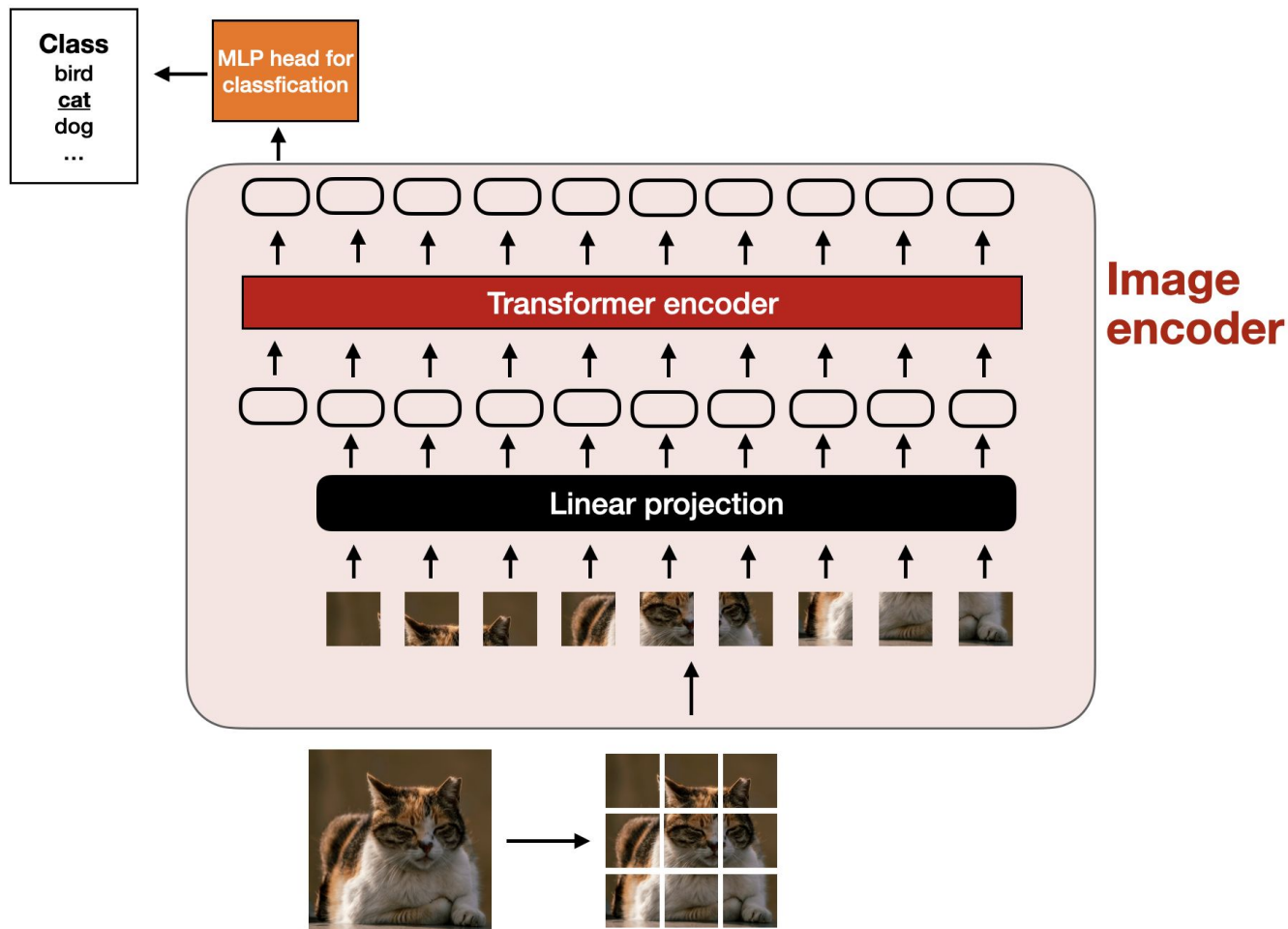
Method A: Unified embedding decoder architecture



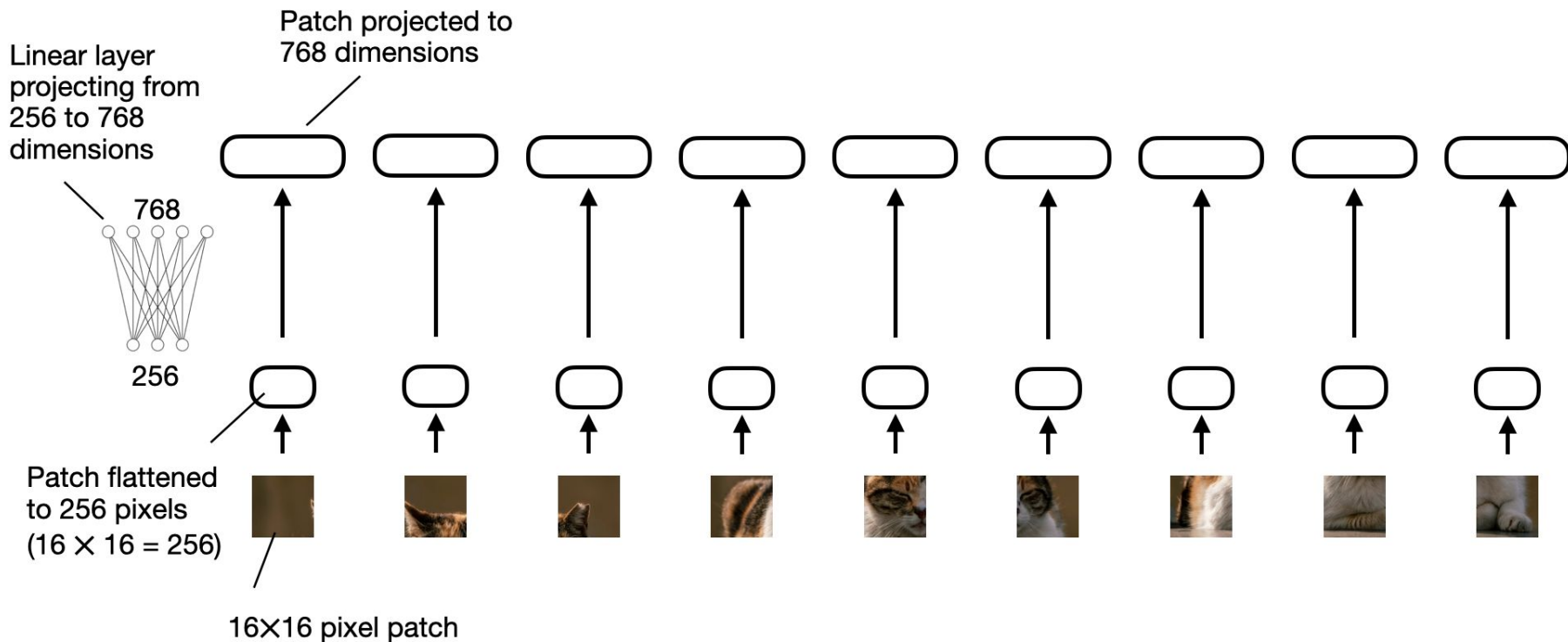
Understanding image encoders



Understanding image encoder (cont'd)

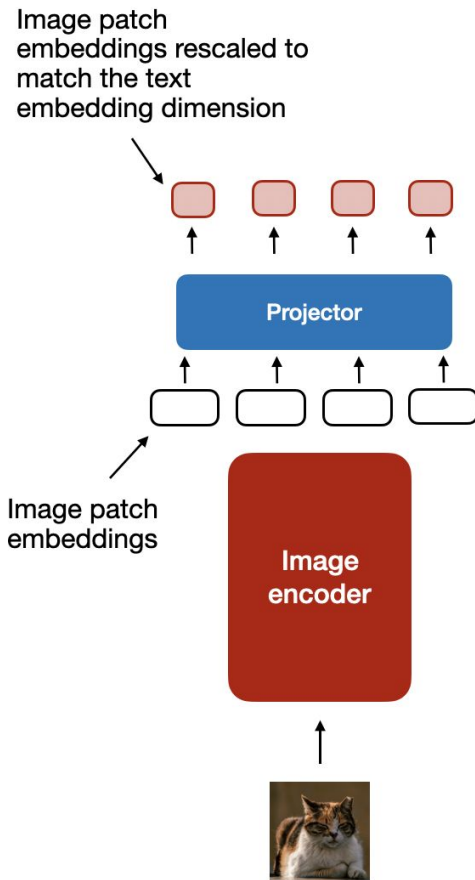


The role of the linear projection module



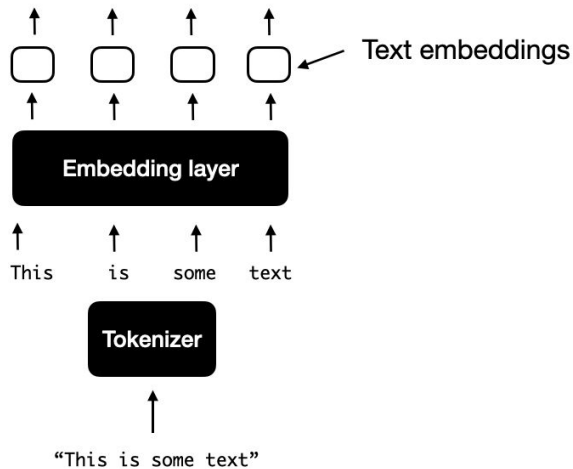
Text and image tokenization and embedding

Image tokenization

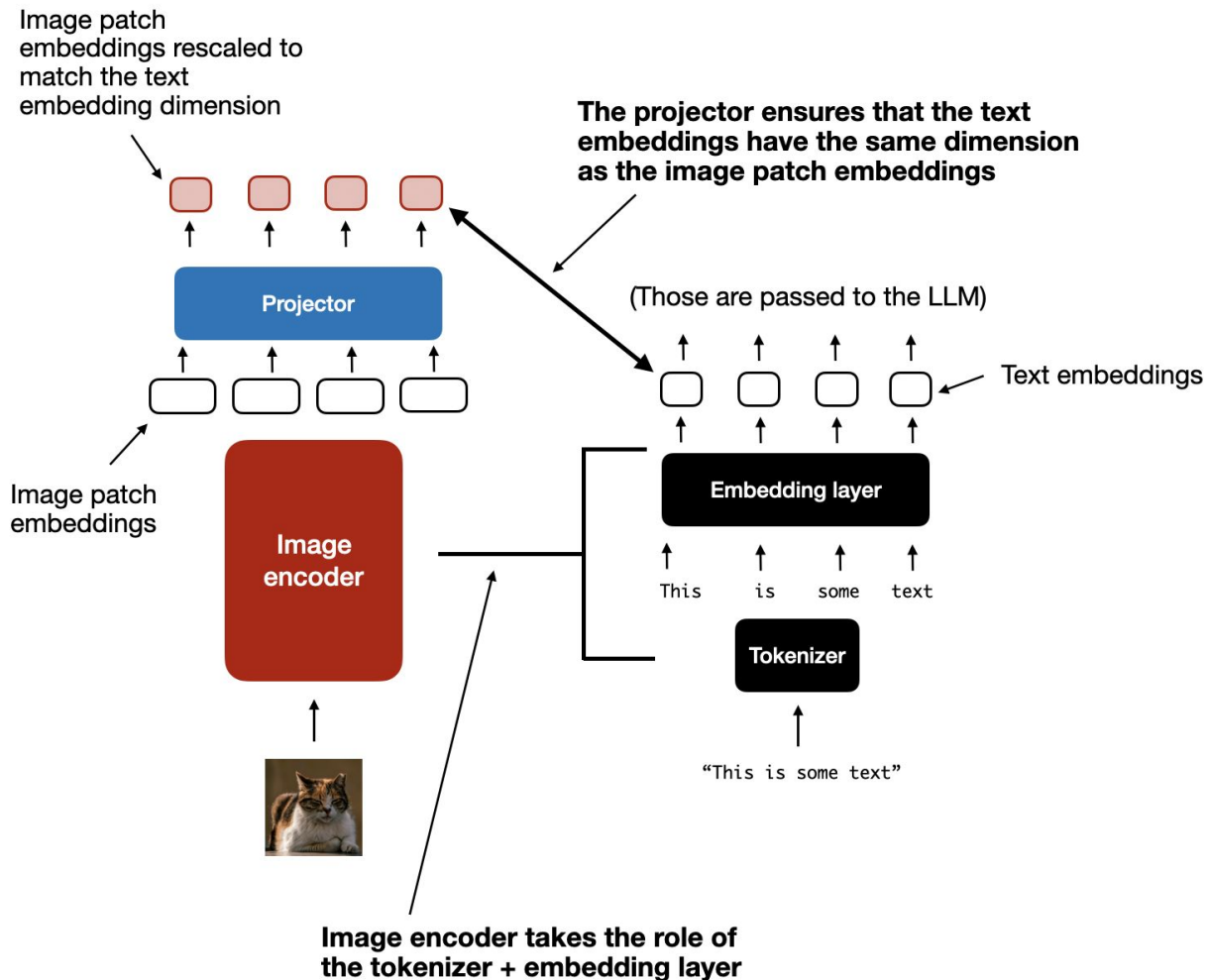


Text tokenization

(Those are passed to the LLM)

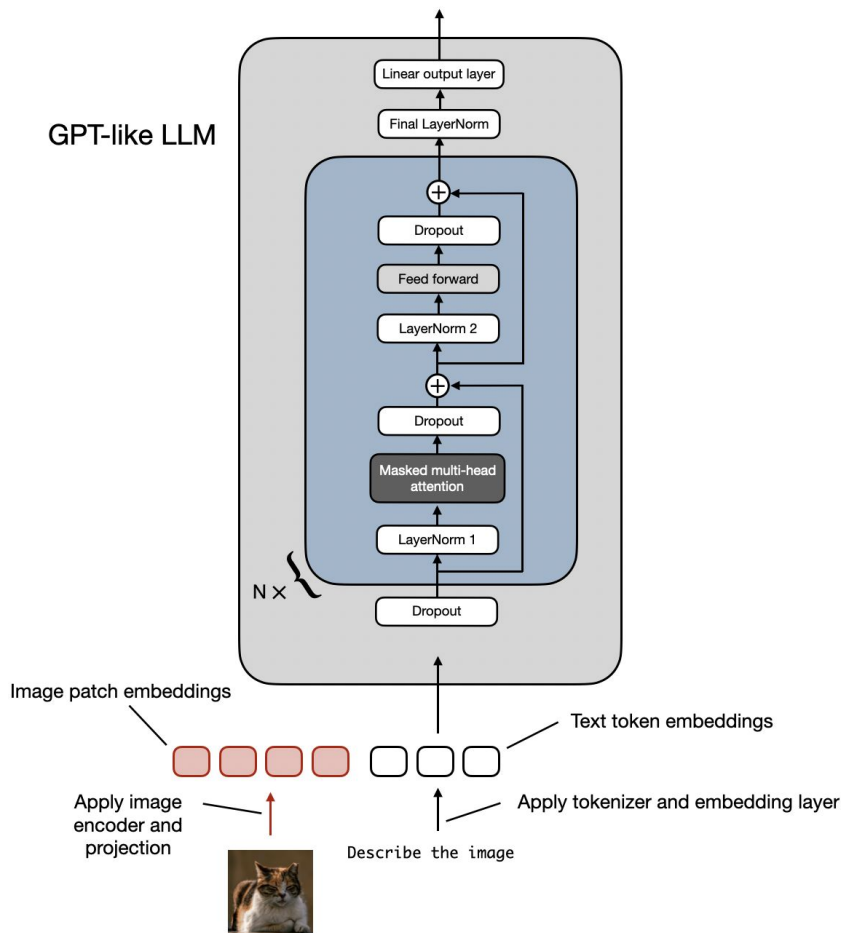


The role of the projector is to match the text token embedding dimensions

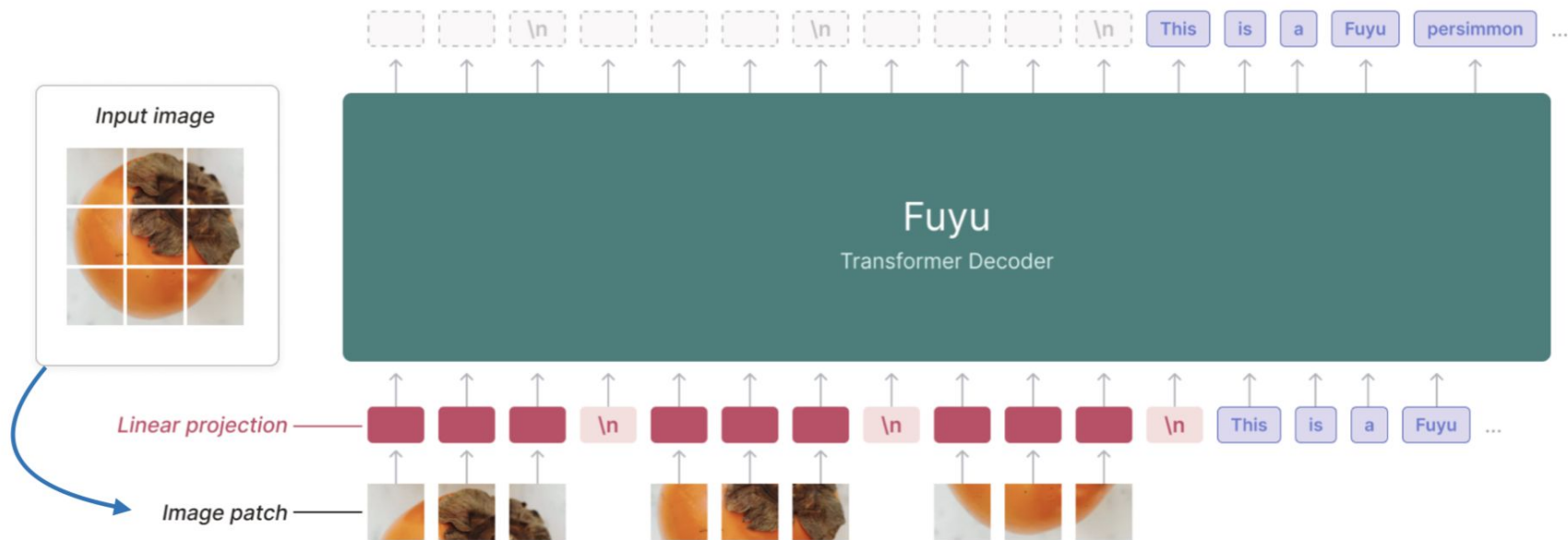


Method A: Unified Embedding Decoder Architecture

We can simply concatenate image and text embeddings



Versions of Method A that operate directly on patches

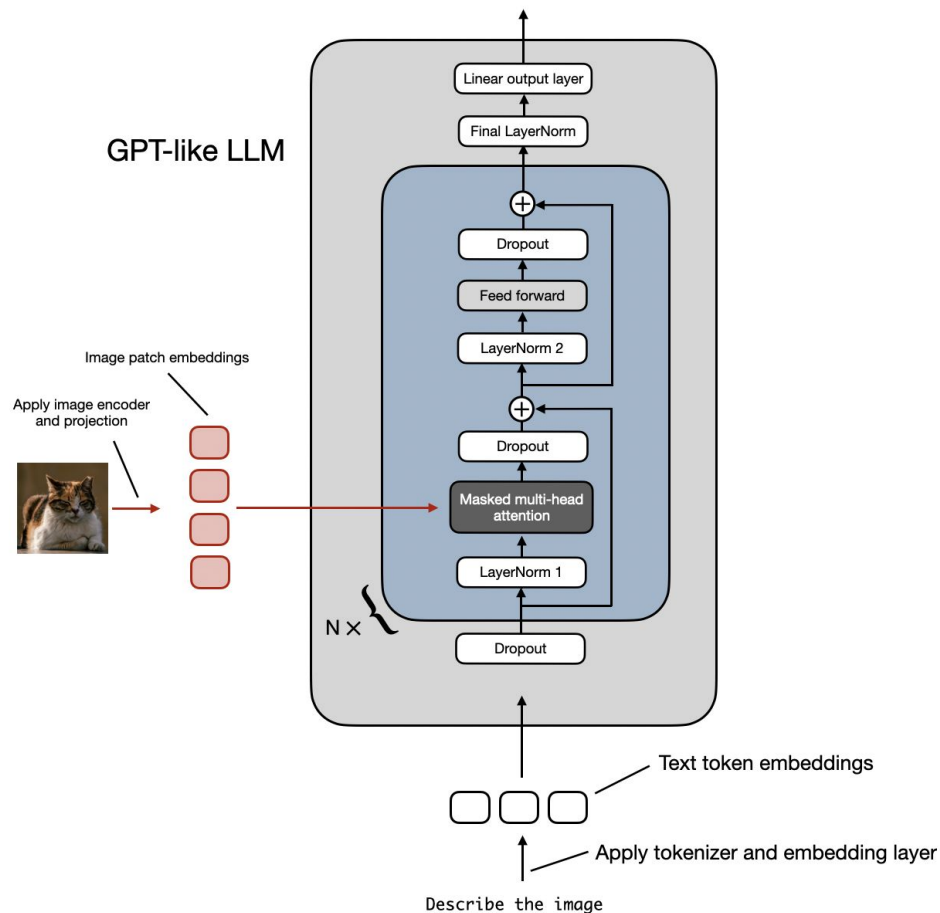


Input image patches are fed directly into an embedding layer

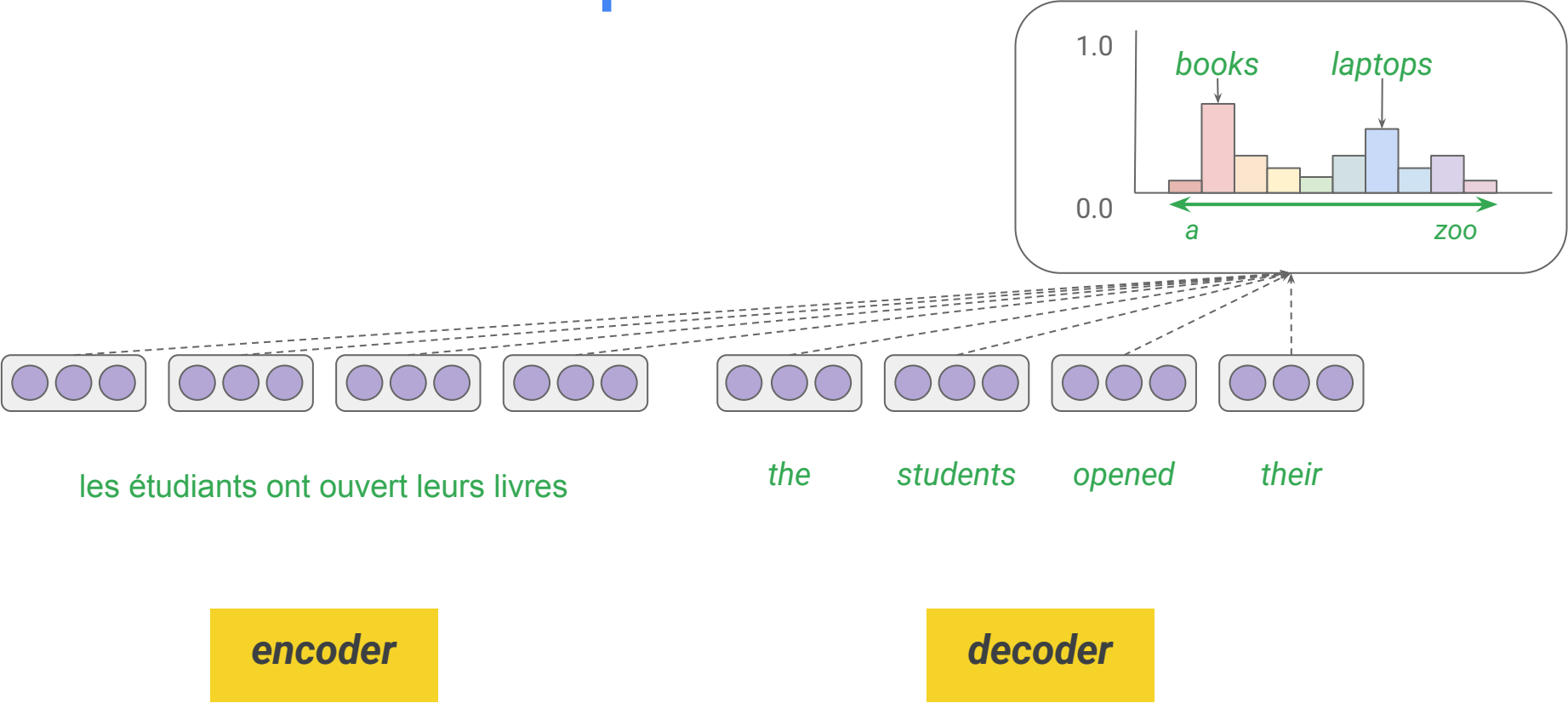
<https://www.adept.ai/blog/fuyu-8b>
Understanding Multimodal LLMs by Sebastian Raschka

Method B: Cross-modality attention architecture

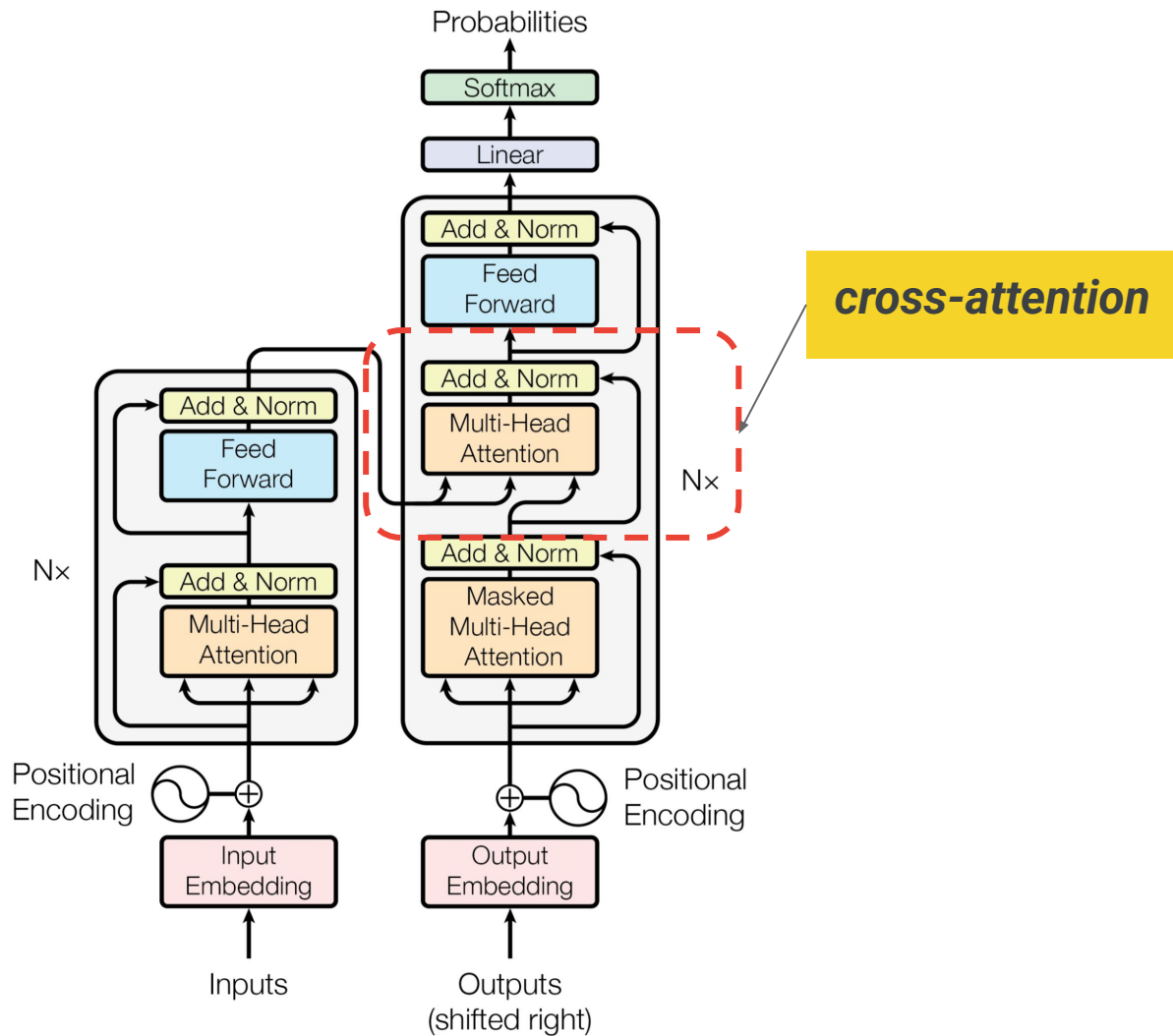
Method B: Cross-Modality Attention Architecture



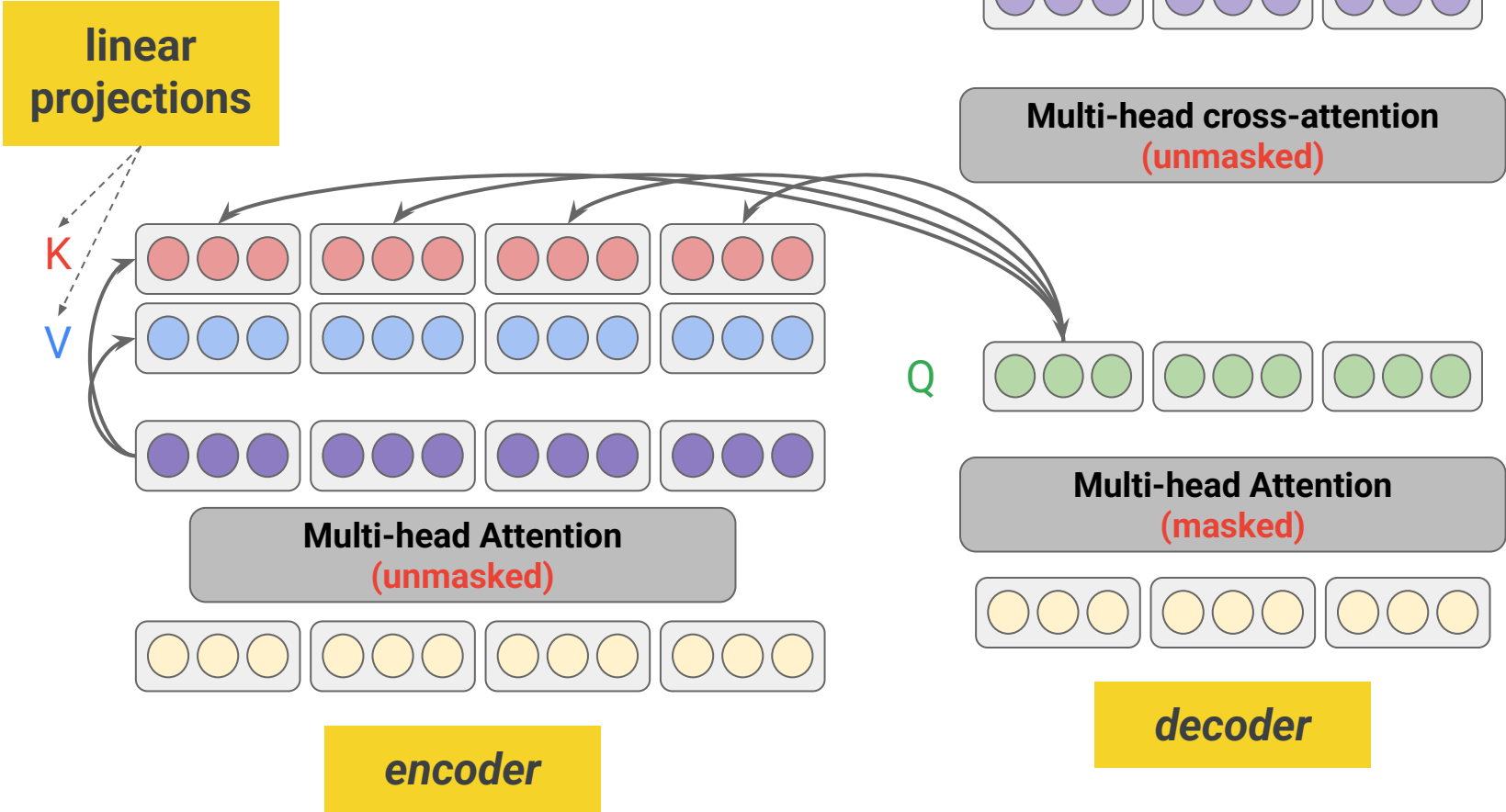
Cross-attention recap



Cross-attention in the decoder



Cross-attention in the decoder



Cross-attention in the decoder (cont'd)

residual
connections

linear
projections

K

V

Multi-head Attention
(unmasked)

encoder

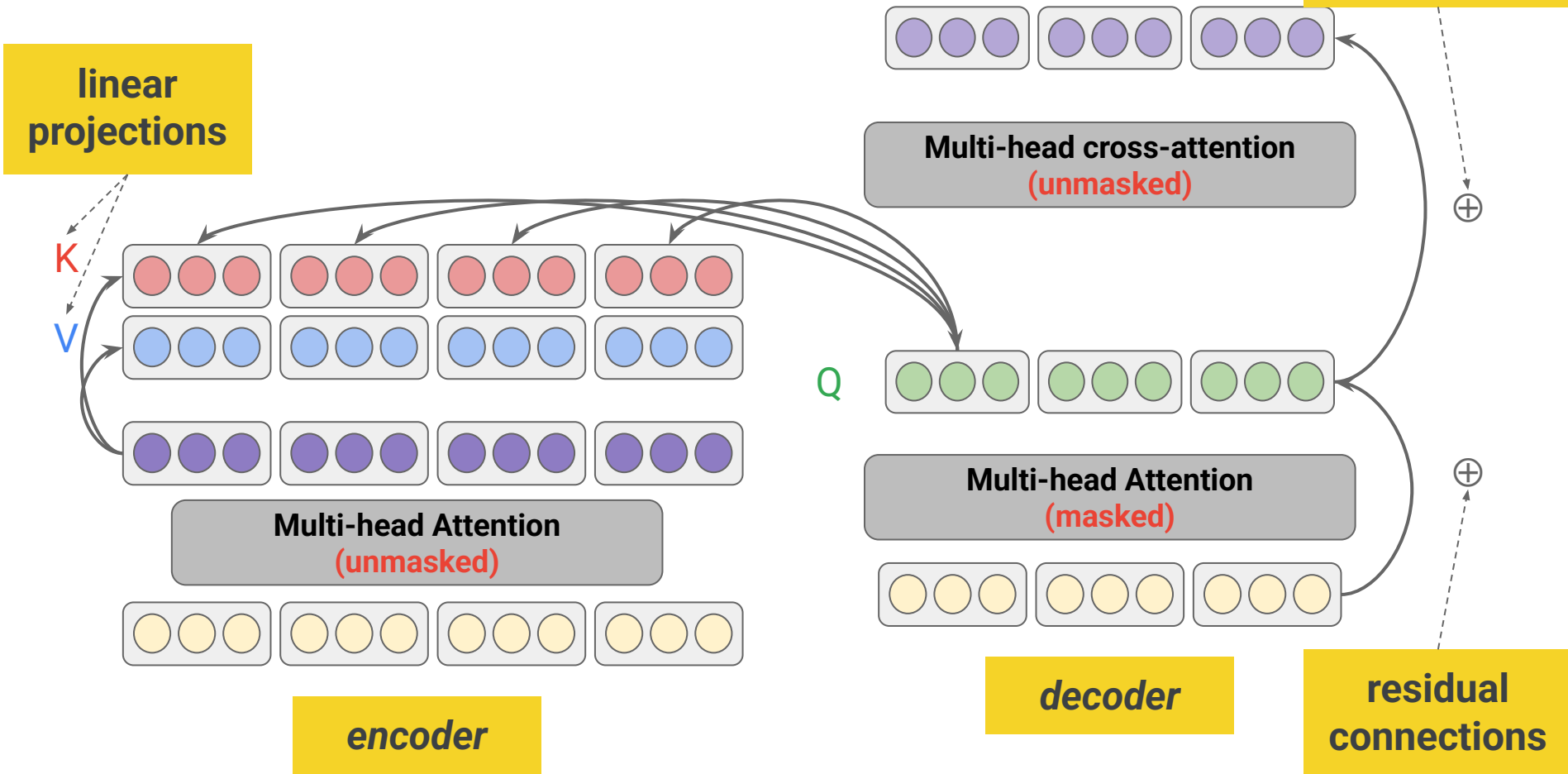
Multi-head cross-attention
(unmasked)

Q

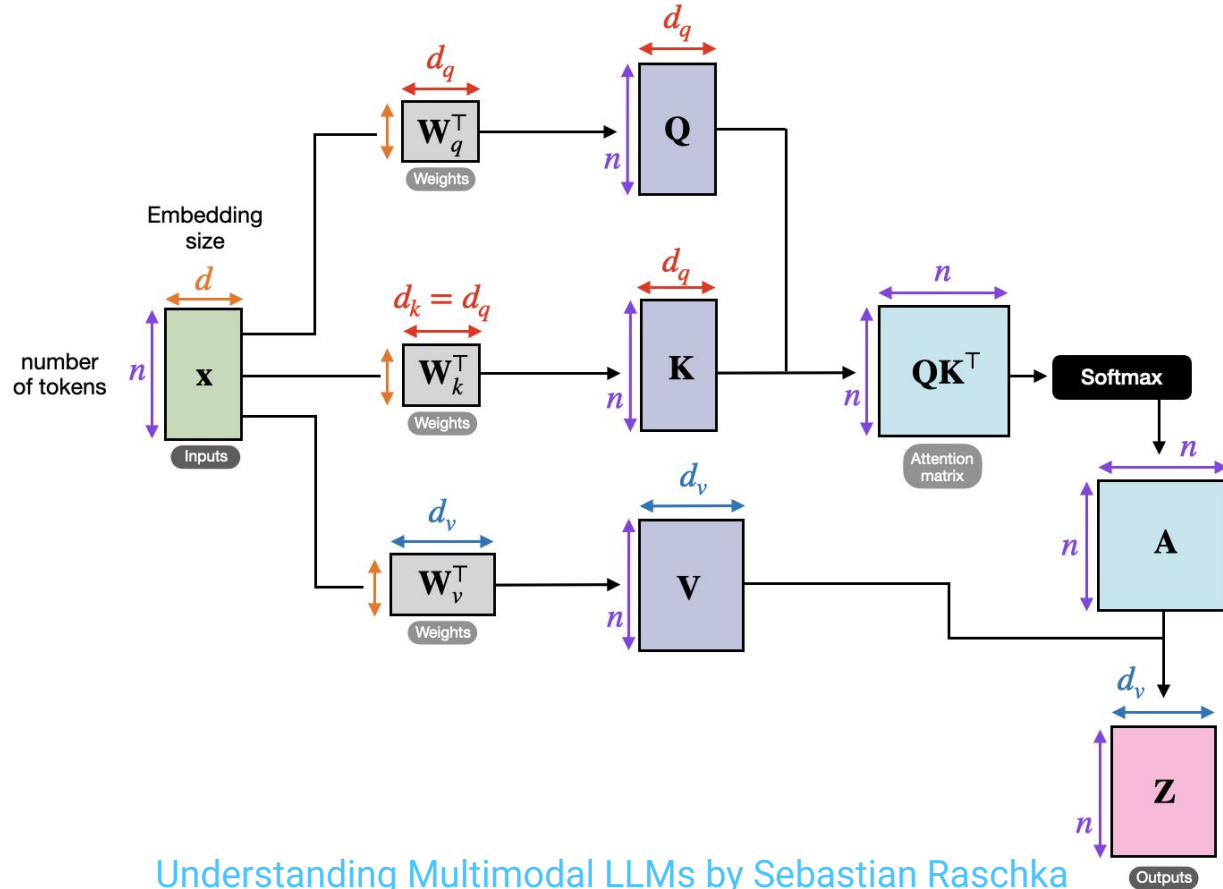
Multi-head Attention
(masked)

decoder

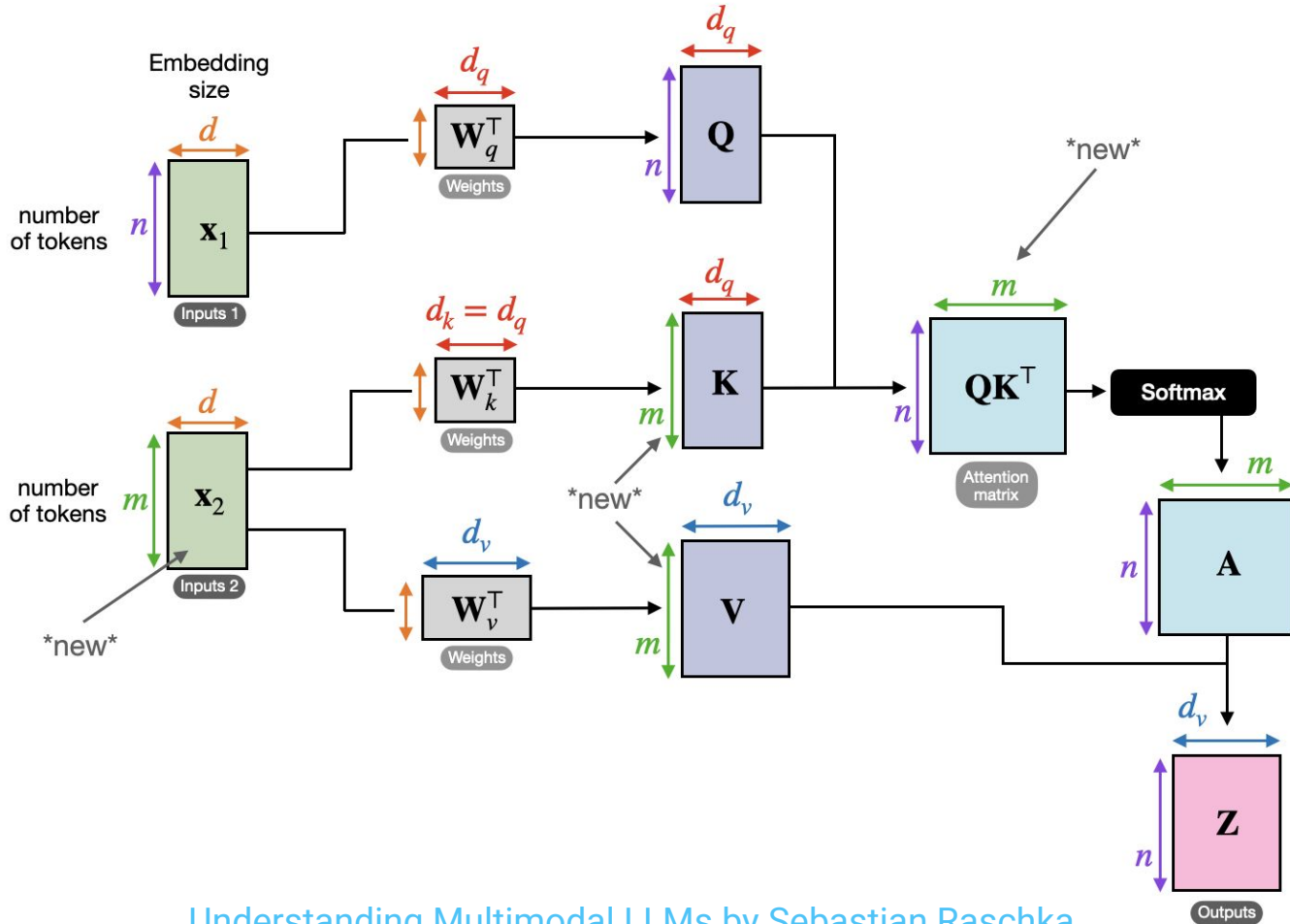
residual
connections



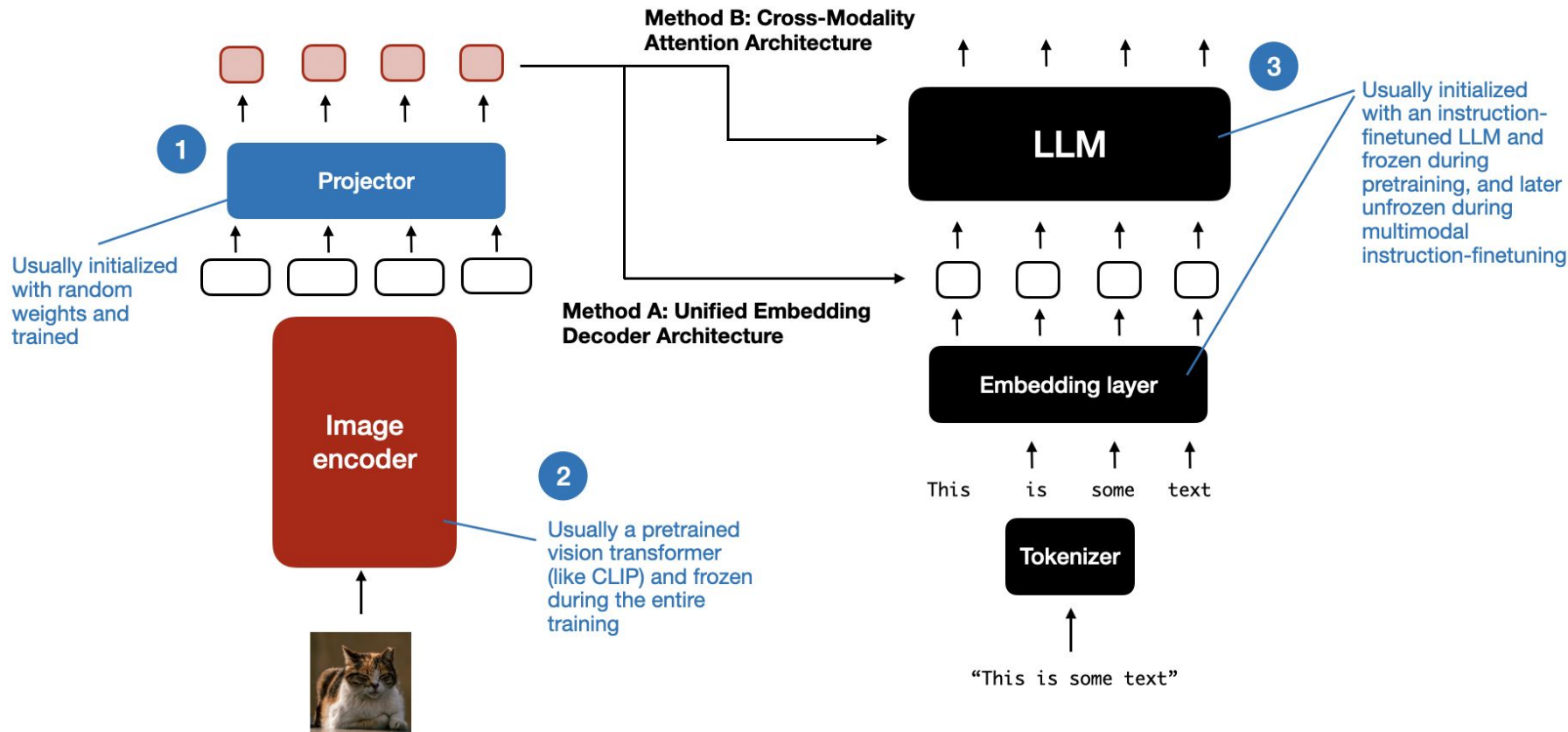
Regular self-attention



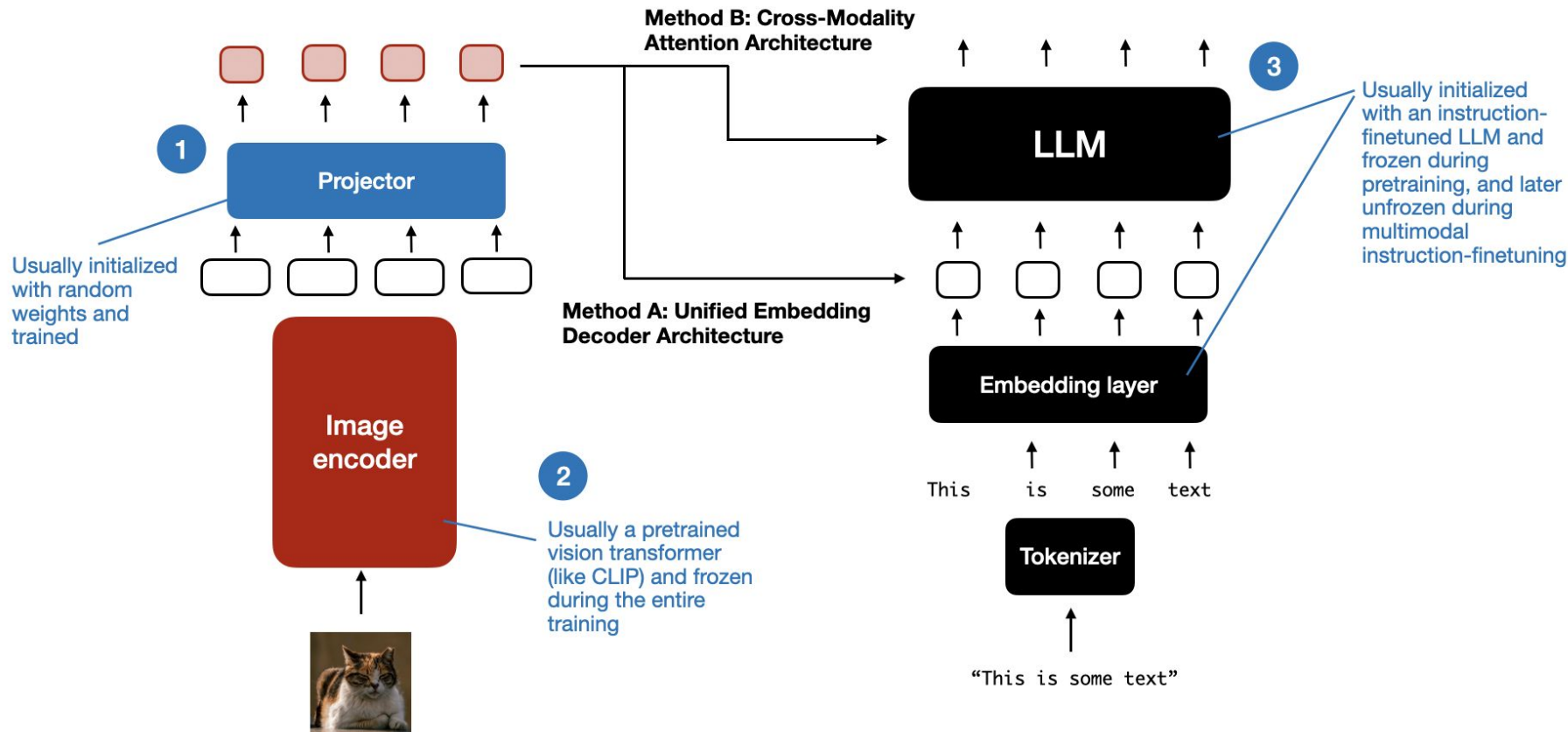
Cross-attention



Unified decoder and cross-attention model training

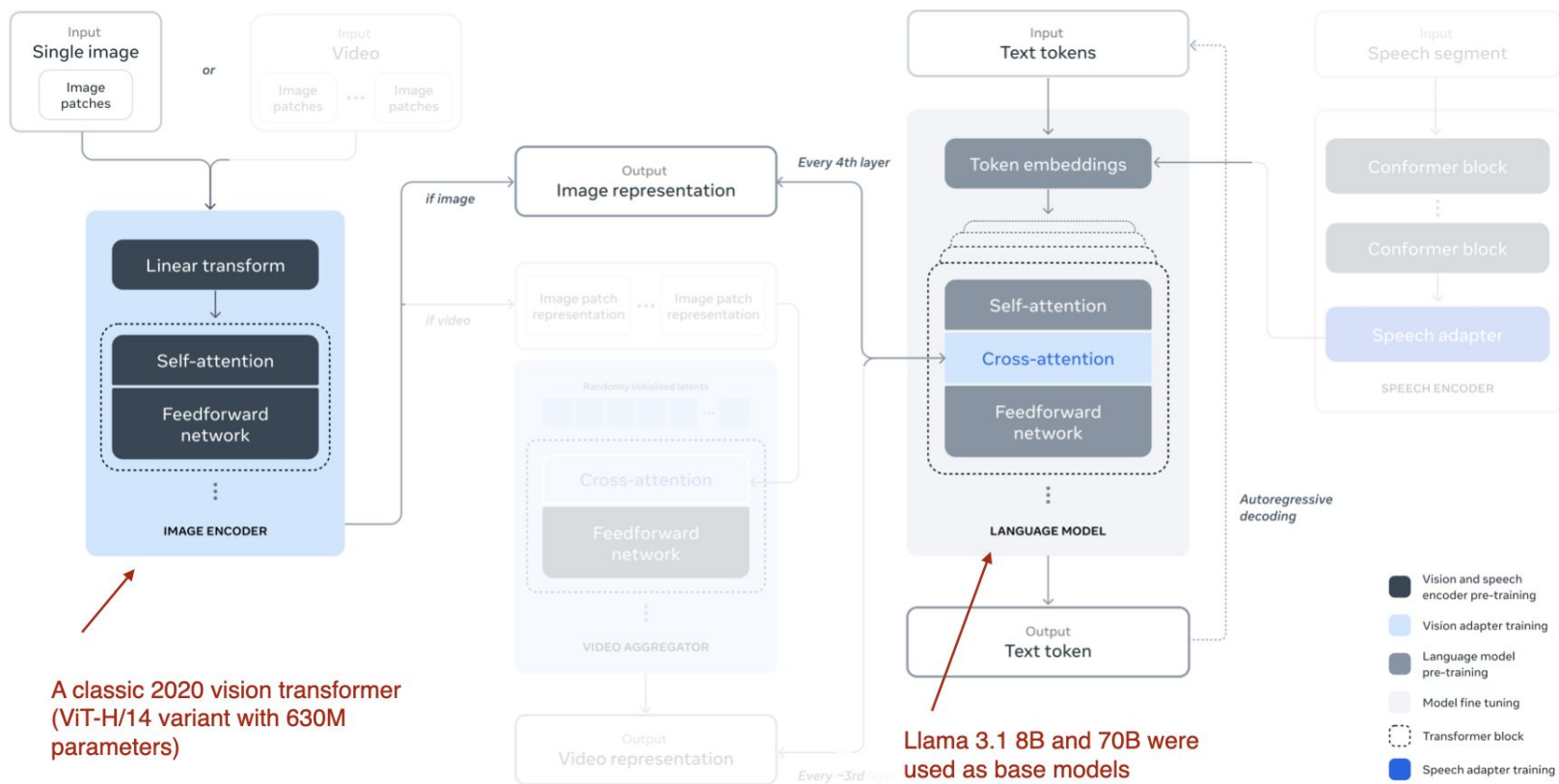


Unified decoder and cross-attention model training



Recent multimodal models and methods

The Llama 3 herd of models



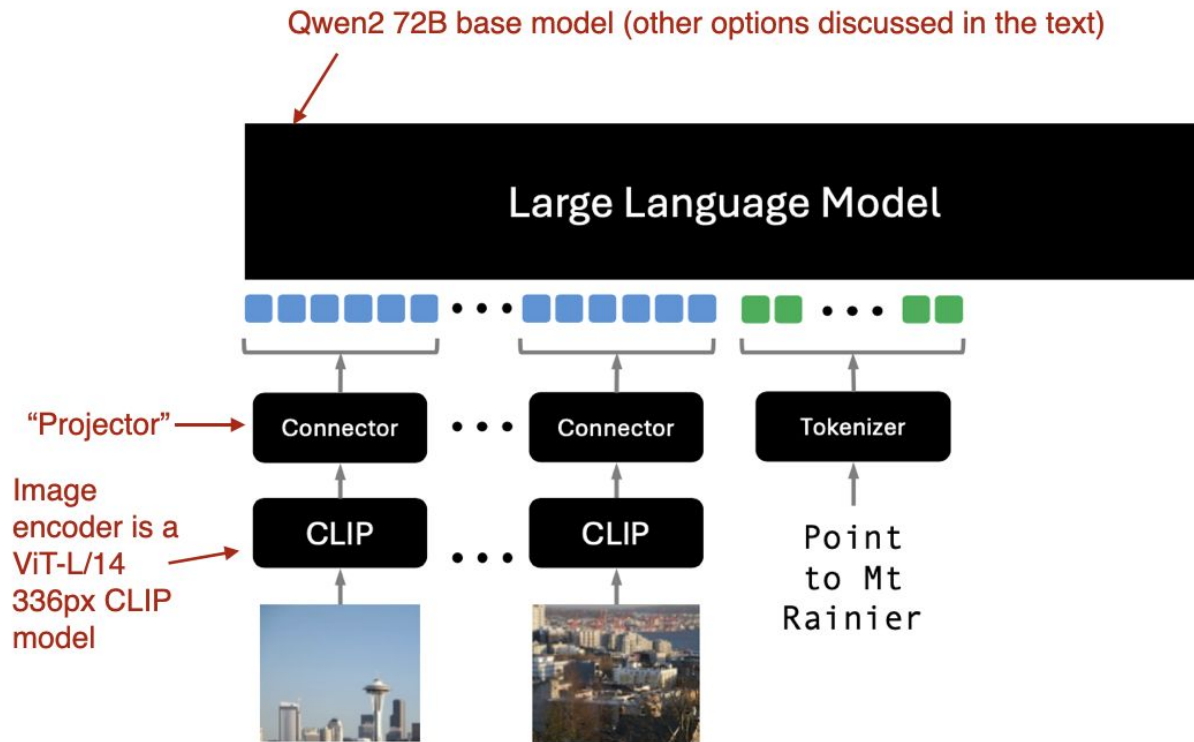
A classic 2020 vision transformer (ViT-H/14 variant with 630M parameters)

Llama 3.1 8B and 70B were used as base models

<https://arxiv.org/abs/2407.21783>

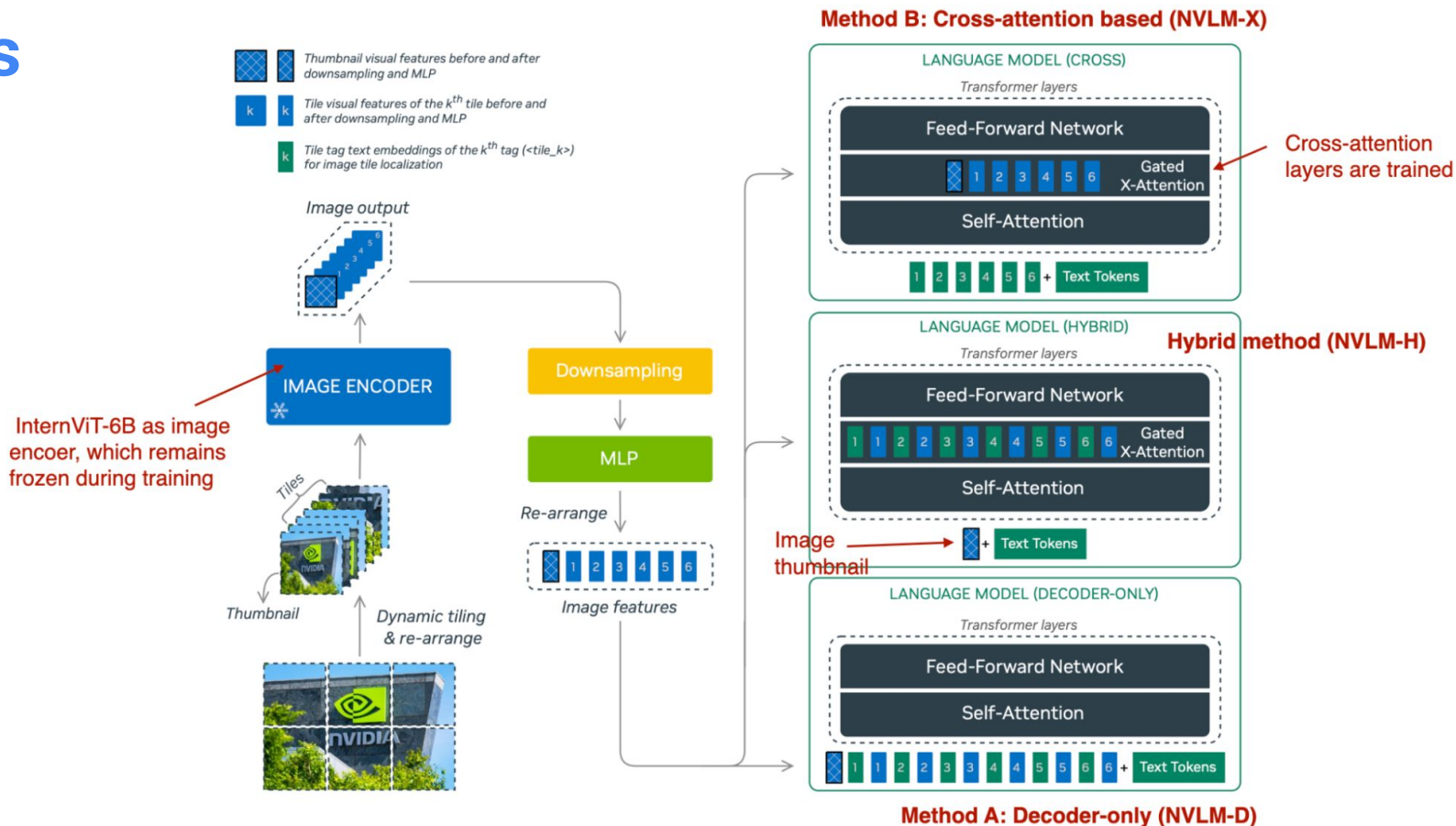
Understanding Multimodal LLMs by Sebastian Raschka

AI2's Molmo and PixMo



<https://www.arxiv.org/abs/2409.17146>
Understanding Multimodal LLMs by Sebastian Raschka

NVIDIA's NVLM



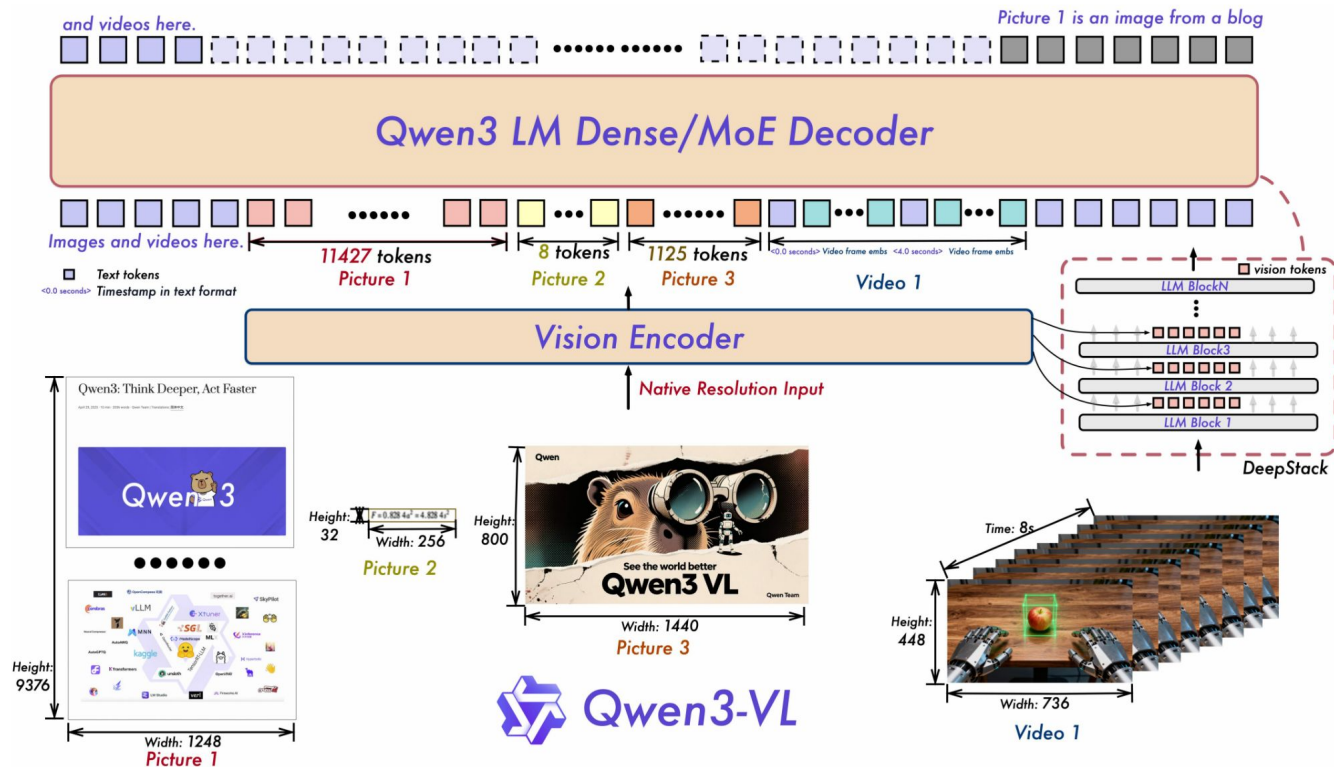
<https://arxiv.org/abs/2409.11402>

Understanding Multimodal LLMs by Sebastian Raschka

NVIDIA's NVLM

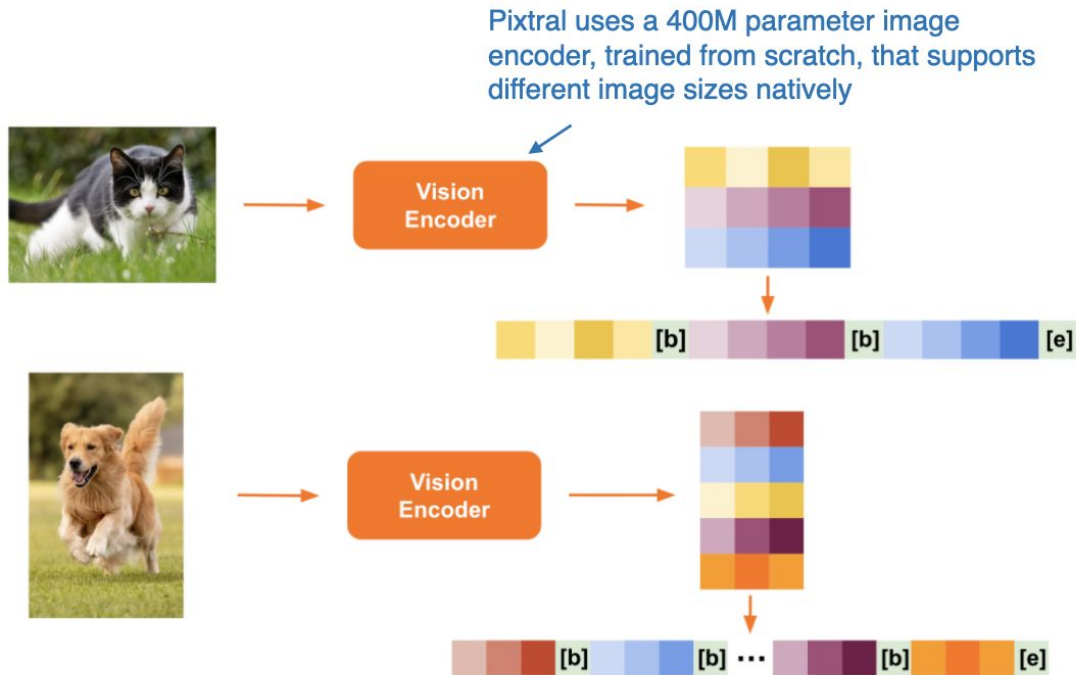
- **NVLM-X** (cross-attention) demonstrates superior computational efficiency for high-resolution images.
- **NVLM-D** (unified embedding) achieves higher accuracy in OCR-related tasks.
- **NVLM-H** combines the advantages of both methods.

Qwen3-VL: Enhancing vision- language model's perception of the world at any resolution



<https://arxiv.org/abs/2511.21631>

Pixtral 12B



<https://mistral.ai/news/pixtral-12b>
[Understanding Multimodal LLMs by Sebastian Raschka](#)

MM1: Methods, analysis & insights

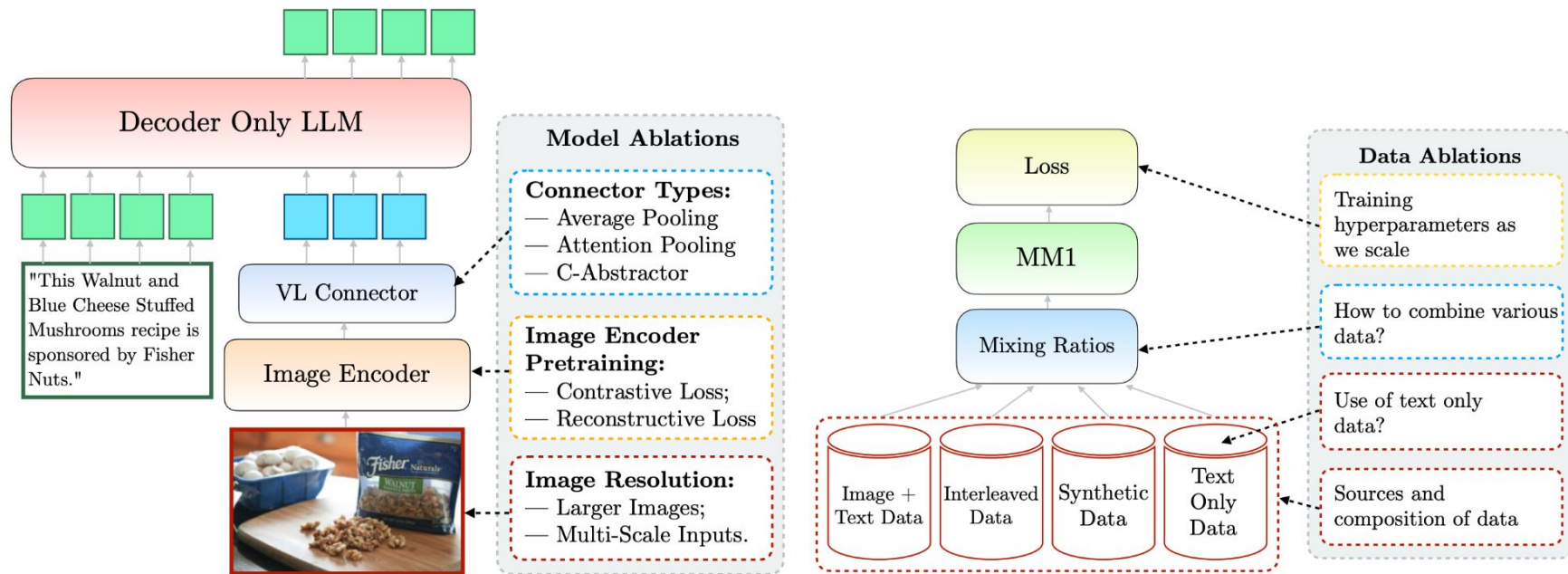
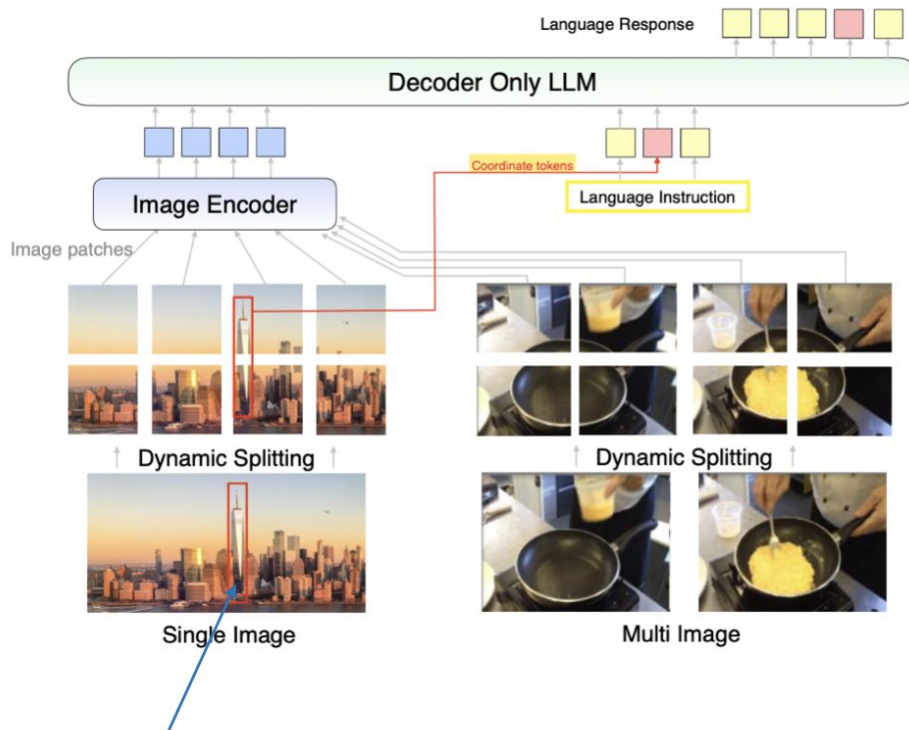


Fig. 3: *Left:* Model ablations: what visual encoder to use, how to feed rich visual data, and how to connect the visual representation to the LLM. *Right:* Data ablations: type of data, and their mixture.

MM1.5: Methods, analysis & insights

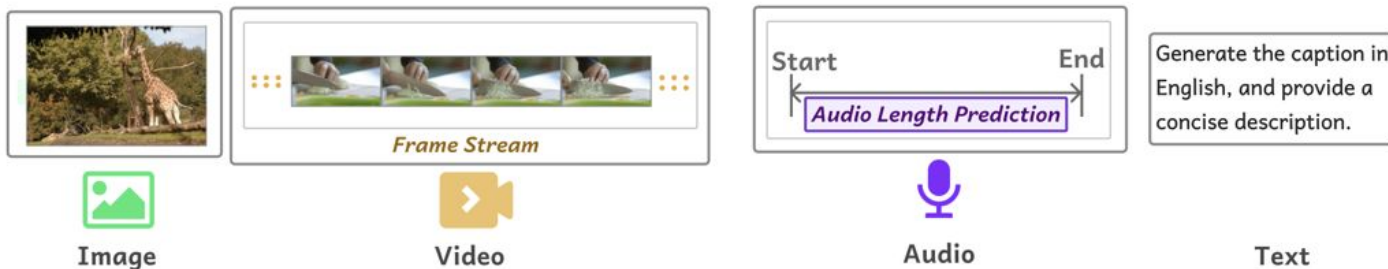
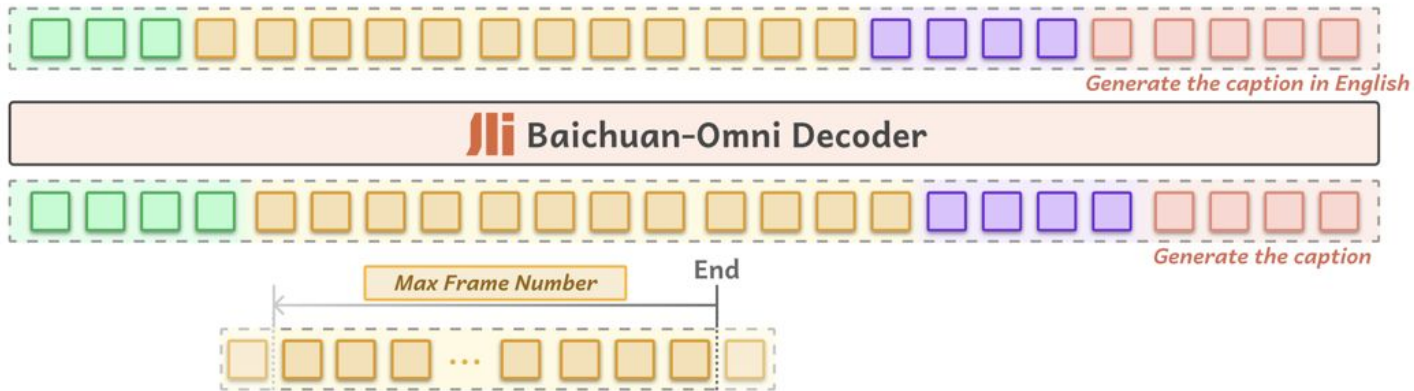


Visual QA data is enriched with bounding boxes (and/or point coordinates)

<https://arxiv.org/abs/2409.20566>
Understanding Multimodal LLMs by Sebastian Raschka

Baichuan-Omni

Baichuan-Omni uses the Unified Embedding
Decoder Architecture setup



<https://arxiv.org/abs/2410.08565>
Understanding Multimodal LLMs by Sebastian Raschka

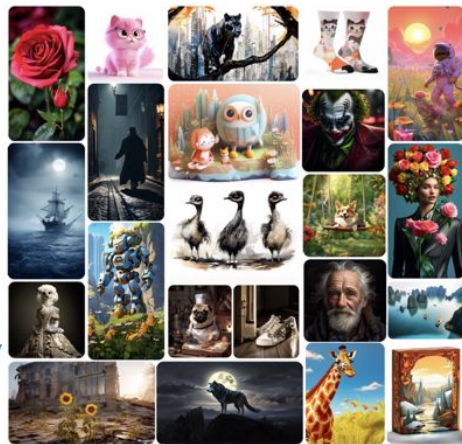
Baichuan-Omni (cont'd)

- **Projector training:** Initially, only the projector is trained, while both the vision encoder and the language model (LLM) remain frozen.
- **Vision encoder training:** Next, the vision encoder is unfrozen and trained, with the LLM still frozen.
- **Full model training:** Finally, the LLM is unfrozen, allowing the entire model to be trained end-to-end.

<https://arxiv.org/abs/2410.08565>

[Understanding Multimodal LLMs by Sebastian Raschka](#)

Emu3: Next-token prediction is all you need



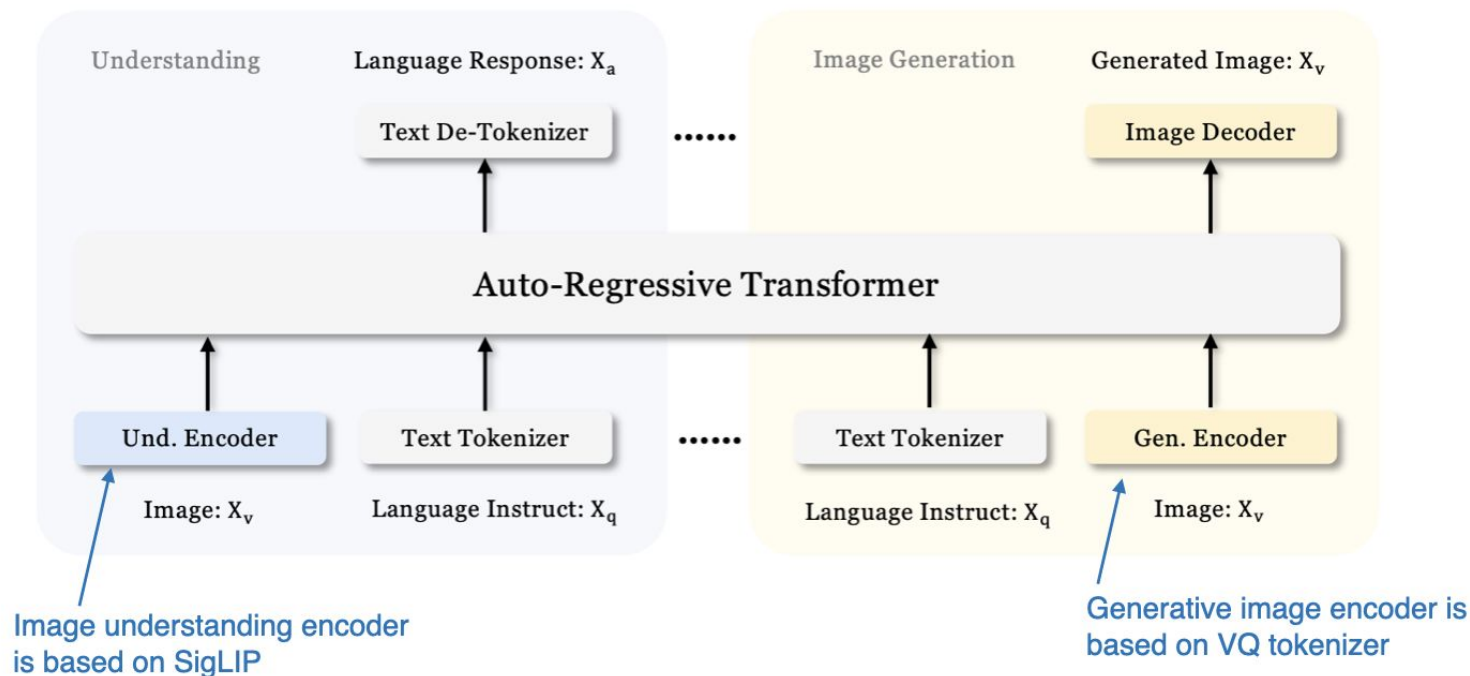
Qualitative results; are said to match or surpass diffusion models like SDXL and DALLÉ-3



<https://arxiv.org/abs/2409.18869>

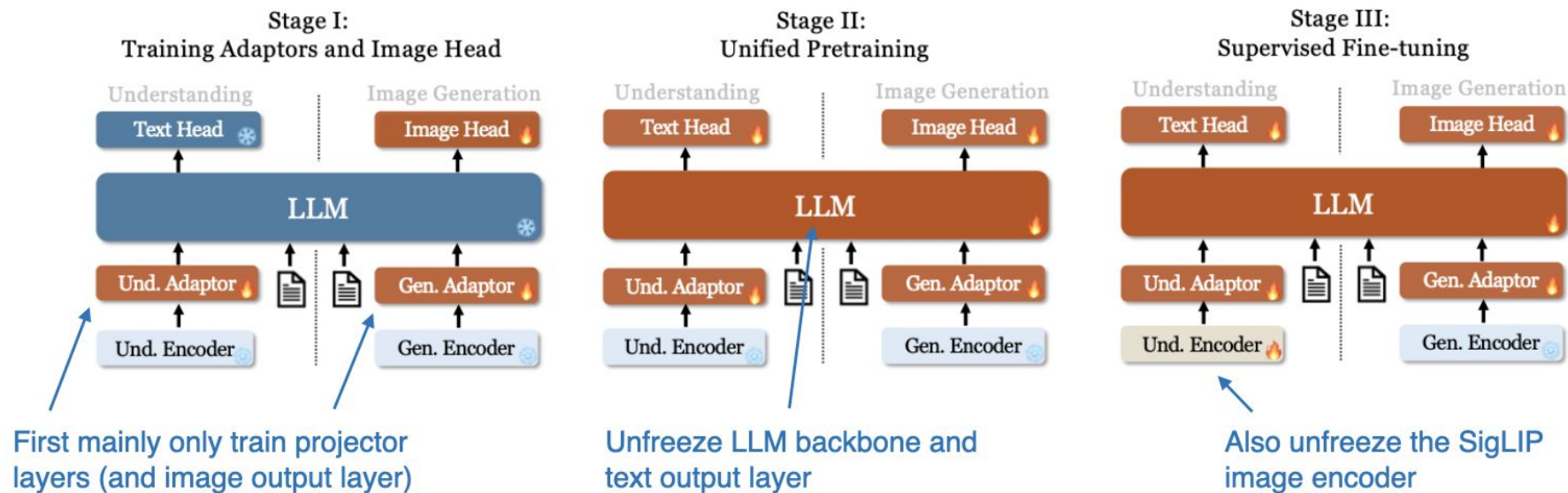
Understanding Multimodal LLMs by Sebastian Raschka

Janus: Decoupling Visual Encoding for Unified Multimodal Understanding and Generation



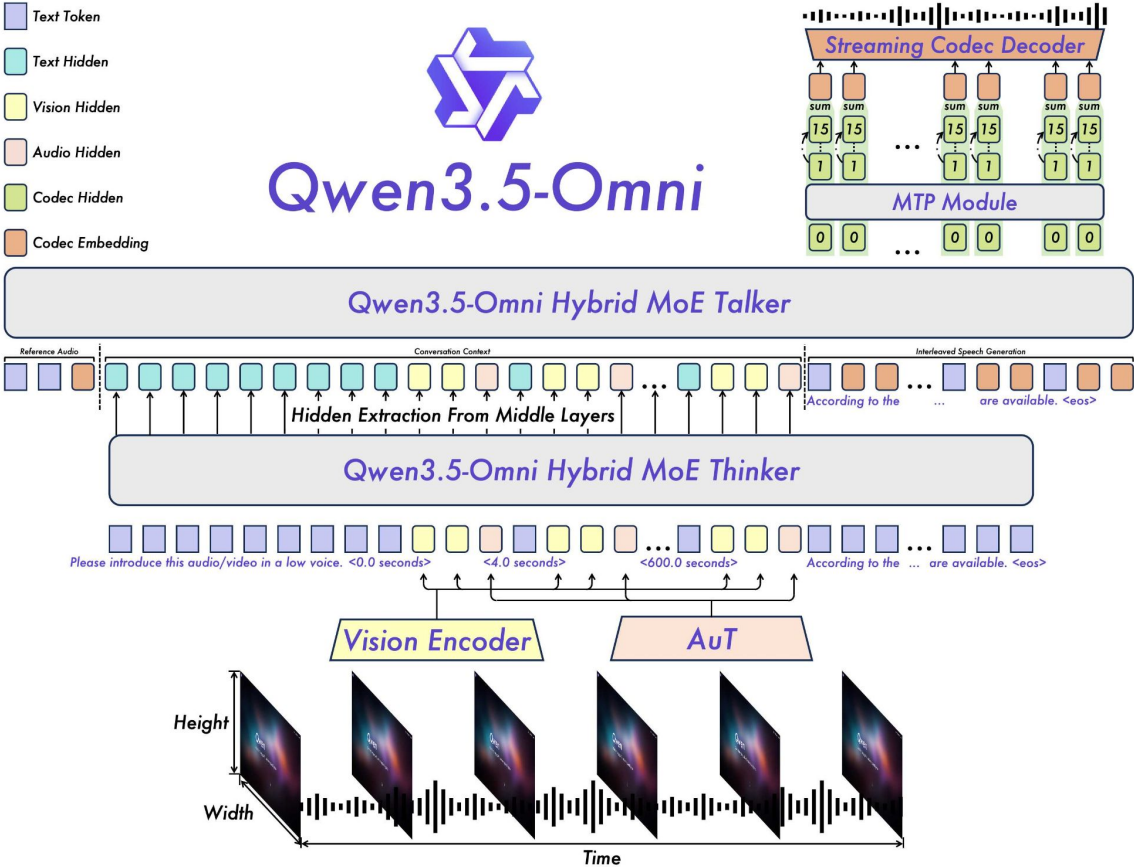
<https://arxiv.org/abs/2410.13848>
Understanding Multimodal LLMs by Sebastian Raschka

Janus: Decoupling Visual Encoding for Unified Multimodal Understanding and Generation



<https://arxiv.org/abs/2410.13848>
Understanding Multimodal LLMs by Sebastian Raschka

Qwen3.5- Omni



GLM-4.5V and GLM-4.1V-Thinking

<|think|>From the given image, ...



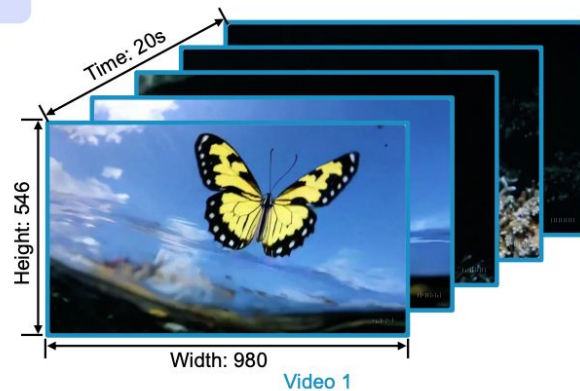
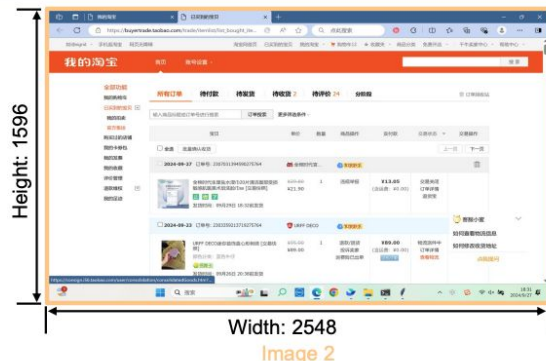
Language Decoder



MLP Projector

ViT Encoder

(Native resolution w. 2x temporal compression)



Thank you!